

Mouse gingival single-cell transcriptomic atlas: An activated fibroblast subpopulation guides oral barrier immunity in periodontitis

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2
November 14, 2023 (this version)

Reviewed preprint version 1
June 2, 2023

Sent for peer review
April 20, 2023

Posted to bioRxiv
April 14, 2023

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Abstract

Periodontitis, one of the most common non-communicable diseases, is characterized by chronic oral inflammation and uncontrolled tooth supporting alveolar bone resorption. Its underlying mechanism to initiate aberrant oral barrier immunity has yet to be delineated. Here, we report a unique fibroblast subpopulation activated to guide oral inflammation (AG fibroblasts) identified in a single-cell RNA sequencing gingival cell atlas constructed from the mouse periodontitis models. AG fibroblasts localized beneath the gingival epithelium and in the cervical periodontal ligament responded to the ligature placement and to the discrete application of Toll-like receptor stimulants to mouse maxillary tissue. The upregulated chemokines and ligands of AG fibroblasts linked to the putative receptors of neutrophils in the early stages of periodontitis. In the established chronic inflammation, neutrophils together with AG fibroblasts appeared to induce type 3 innate lymphoid cells (ILC3s) that were the primary source of interleukin-17 cytokines. The comparative analysis of *Rag2*^{-/-} and *Rag2^{yc}*^{-/-} mice suggested that ILC3 contributed to the cervical alveolar bone resorption interfacing the gingival inflammation. We propose that AG fibroblasts function as a previously unrecognized surveillant to initiate gingival inflammation leading to periodontitis through the AG fibroblast-neutrophil-ILC3 axis.

eLife assessment

The findings of this article provide **valuable** information on the changes of cell clusters induced by chronic periodontitis. The observation of a new fibroblast subpopulation, named AG fibroblasts, is interesting, and the strength of evidence presented is **solid**.

The oral mucosa is one of the most active barrier tissues in the human body [1, 2] and oral barrier immunity acts as a crucial surveillance system, which must defend from pathogenic organisms, while ensuring sustained tolerance of commensal microbes and harmless antigens [3]. However, once this gingival inflammation progresses, adjacent alveolar bone supporting the cervical area of the dentition is subjected to a localized, but often severe, osteoclastic bone resorption, resulting in tooth loss and disruption of the maxillofacial structure [4]. Periodontitis is not only the most frequent cause of tooth loss in adults [5], but also, globally, ranks among the most significant contributors to poor health and decreased quality of life, imposing substantial economic and healthcare burdens [6]. It is therefore acutely important to elucidate the molecular mechanisms underlying the initiation of gingival inflammation, in order to facilitate the development of effective preventive and therapeutic measures for this debilitating condition.

The postulated pathological framework has been reconstructed from animal models [7, 8], harvested immune cell analyses, and oral microbial associations [4, 9, 10]. For example, the ligature-induced periodontitis mouse model has shown that fully developed periodontal inflammation is closely associated with the abundant recruitment and excessive activation of neutrophils [11] and pathological induction of interleukin (IL)-17-secreting proinflammatory effector CD4⁺ T helper (Th)17 cells [12]. Based on our current understanding of osteoimmunology, it was hypothesized that these pathological immune cells activate osteoclasts, leading to induction of periodontal alveolar bone resorption [13, 14]. However, the precise cascade of events leading to this disease outcome has not been fully deciphered.

Investigation of the molecular mechanisms underlying interaction between different inflammatory cell fractions in immunological disorders is an intense area of research. Prior studies have shown that chemokines are involved in the trafficking and activation of inflammatory cells during both homeostasis and disease-associated inflammation [14]. These small proteins, which typically contain disulfide cysteine–cysteine (CC) and cysteine–X–cysteine (CXC) molecular signatures, act as ligands for cellular receptors that modulate inflammatory signaling pathways [15]. Such chemokine–receptor networks direct the migration of inflammatory cells, potentially amplifying the resulting tissue damage. However, increasing evidence suggests that in addition to immune cells, barrier tissue stromal cells are involved in innate immune cell regulation [16–19]. We and others have found that oral fibroblasts secrete cytokines and chemokines in response to microbial stimuli or a proinflammatory environment, suggesting a possible mechanism by which they may regulate immune cell function [8, 20]. Therefore, a comprehensive elucidation of chemokine–receptor signaling networks will likely need to include all types of gingival cells.

In the present study, to better understand the pathways contributing to chronic oral inflammation, we constructed the gingival single-cell transcriptomic atlas of the ligature-induced periodontitis mouse model as well as the newly developed mouse model of maxillary topical application (MTA) of pathological oral microbial biofilm component(s), discretely mimicking the initial events of gingival inflammation and degradation. Here we report a novel subpopulation of fibroblasts activated to guide chronic inflammation (AG fibroblasts). Our findings suggest that chemokines and ligands derived from AG fibroblasts bind to and activate receptors involved in neutrophil and lymphocytes. We further show that in the fully developed chronic gingival inflammation of the ligature-induced periodontitis model, Cd4⁺ innate lymphoid cells (ILCs) predominantly produce proinflammatory IL-17 cytokines and provide evidence that type 3 ILCs (ILC3s) are responsible for cervical alveolar bone resorption in mice—a pathological feature consistent with the human periodontitis disease phenotype. Thus, the results from this study identify the AG fibroblast–neutrophil–ILC3 axis as a previously unrecognized mechanism contributing to pathological inflammation in periodontitis.

Alterations in major cell type proportions during periodontitis development

The alveolar bone and gingival tissue surrounding the maxillary second molar of C57BL/6J mice presented healthy connective tissue supporting the dentition with minimal CD45+ immune cell infiltration and well-developed collagen architecture (**Figure 1A** [↗](#) – **figure supplement 1A** [↗](#)). The placement of a ligature around the maxillary left second molar [\[21](#) [↗](#)] induced a localized and small gingival connective tissue degradation and infiltration of CD45+ immune cells near the cervical area (**Figure 1B** [↗](#)). The noticeable gingival defect and more prominent though still localized CD45+ immune cell infiltration developed after Day 3 following ligature placement (**Figure 1C** [↗](#) – **figure supplement 1A** [↗](#), **1B** [↗](#)). On Day 7, the gingival connective tissue was largely degraded, and chronic inflammation was developed (**Figure 1D** [↗](#) – **figure supplement 1A** [↗](#), **1B** [↗](#)). Three-dimensional (3D) reconstruction of micro-computed tomography (microCT) images (**figure supplement 1C** [↗](#)) further revealed a reduction in alveolar bone height starting from Day 3, which progressively increased on Day 7 (**figure supplement 1D** [↗](#), **1E** [↗](#)). Overall, the observed pattern of periodontal tissue degradation was consistent with that reported in previous studies [\[22](#) [↗](#), [23](#) [↗](#)].

Left-side palatal gingiva tissue was harvested from mice on Day 0 (*i.e.*, healthy gingiva without ligature placement) and on Days 1, 4, and 7 after ligature placement, and gingival cells were dissociated for single-cell RNA sequencing (scRNA-seq) [\[8](#) [↗](#), [24](#) [↗](#)]. On Days 0 and 1, the major cell types identified included epithelial cells expressing cadherin 1 (*Cdh1*) [\[25](#) [↗](#)] and type XVII collagen (*Col17a1*) [\[26](#) [↗](#)], fibroblasts expressing type I collagen (*Col1a1*) [\[27](#) [↗](#)] and lumican (*Lum*) [\[28](#) [↗](#)], B cells expressing membrane spanning 4 domains A1 (*Ms4a1*) [\[29](#) [↗](#)] and cluster of differentiation 79A (*Cd79a*) [\[30](#) [↗](#)], T cells expressing epsilon subunit of T cell receptor complex (*Cd3e*) [\[31](#) [↗](#)] and cluster of differentiation 5 (*Cd5*) [\[32](#) [↗](#)], and myeloid cells expressing lysozyme 2 (*Lyz2*) [\[33](#) [↗](#)] and integrin subunit alpha M (*Itgam*) [\[34](#) [↗](#)] (**Figure 1E and F** [↗](#)—**figure supplement 2** [↗](#)). On Days 4 and 7, an additional endothelial cell fraction expressing selectin P (*Selp*) [\[35](#) [↗](#)] and selectin E (*Sele*) [\[36](#) [↗](#)] emerged (**Figure 1E and F** [↗](#)—**figure supplement 2** [↗](#)), suggesting increased inflammatory neovascularization with the progression of periodontal inflammation. The proportion of B cells was increased on Day 1, and the proportion of myeloid cells increased progressively from Day 4 to 7 (**Figure 1F** [↗](#)). In addition, the proportion of fibroblasts was increased on Day 7 (**Figure 1F** [↗](#)).

Fibroblasts activated to guide leukocyte migration in periodontitis development

We previously identified two distinct subpopulations of gingival fibroblasts, differentiated by expression of type XIV collagen (*Col14a1*) [\[8](#) [↗](#)], and these were also detected in our current scRNA-seq data from Day 0 to 7 (**Figure 2A** [↗](#)). Gene ontology (GO) enrichment analysis of the *Col14a1*-expressing fibroblast subpopulation revealed expression of major gene clusters related to immune regulation, including “Regulation of leukocyte migration” (**Figure 2B** [↗](#)). This immune regulatory phenotype appears to be unique to *Col14a1*-expressing fibroblasts, which are therefore referred to as ‘fibroblasts activated to guide leukocyte migration’ or AG fibroblasts. The other subpopulation of *Col14a1*-expressing fibroblasts appeared to keep typical fibroblastic features and is thus referred to as KT fibroblasts (**Figure 2C** [↗](#)). GO analysis suggested that KT fibroblasts primarily maintained connective tissue remodeling but did not express an immune regulatory phenotype. An additional fibroblast subpopulation expressing smooth muscle aortic actin 2 (*Acta2*) was detected on Day 7, and these were identified as myofibroblasts [\[37](#) [↗](#)] (MF) (**Figure 2A** [↗](#)). We found that the proportions of KT and AG fibroblasts were equal on Day 0 and on Days 1 and 4 after ligature placement (**Figure 2E** [↗](#), **2F** [↗](#)). However, on Day 7, the proportion of AG fibroblasts decreased, and the MF fraction emerged (**Figure 2E** [↗](#), **2F** [↗](#)).

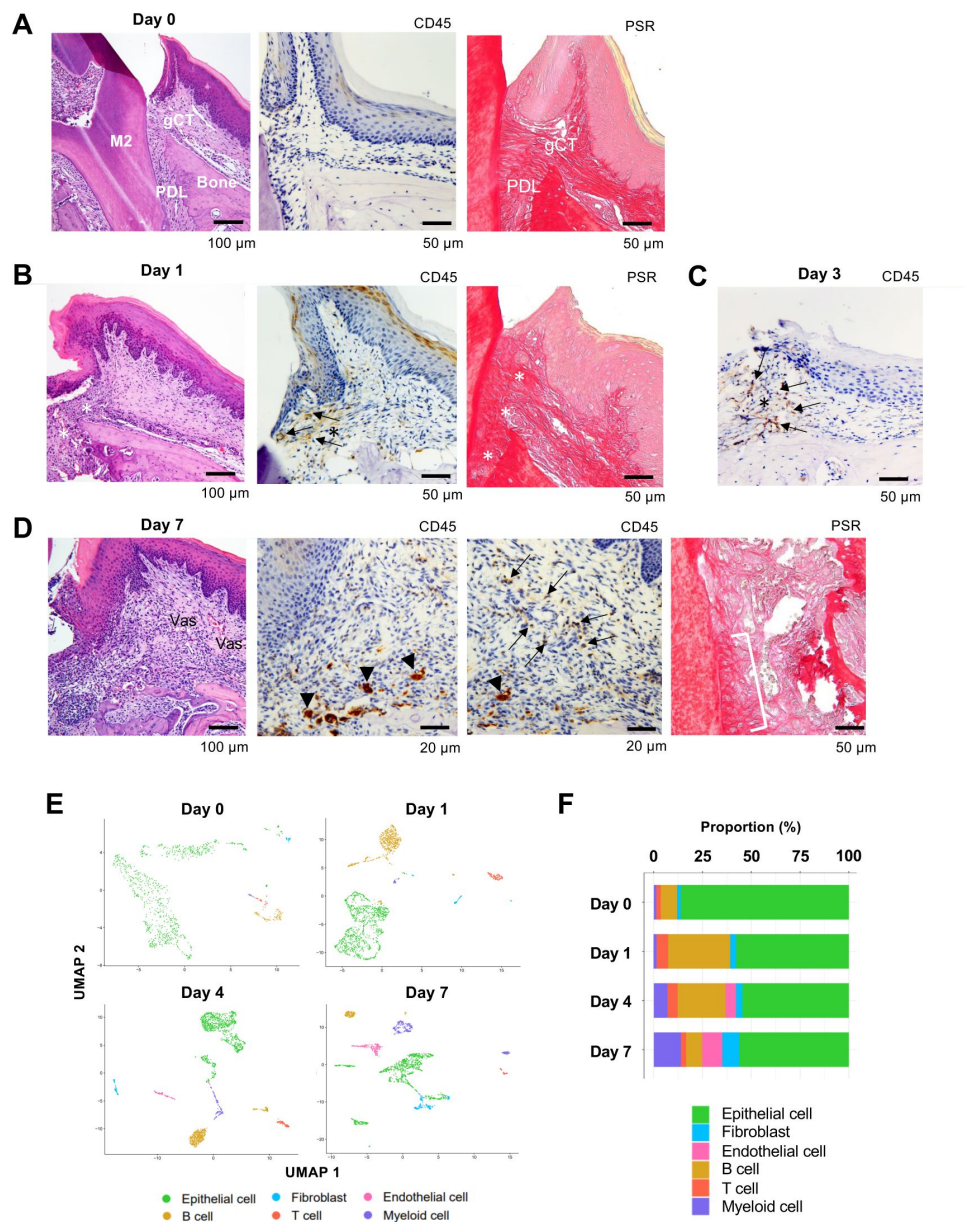


Figure 1.

Changes in proportions of major cell types during ligature-induced periodontitis development in mice. **(A)** Mouse maxillary second molar (M2) has multi-roots and supported by alveolar bone (Bone), periodontal ligament (PDL) and gingiva connective tissue (gCT). The oral barrier immunity is not constitutively activated as evidenced by the lack of CD45+ immune cells. Picrosirius red (PSR) stained collagen fibers connected the root surface and alveolar bone in the PDL and organized as dense parallel bundles in gCT. **(B)** One day (Day 1) after a ligature (5.0 silk suture) was placed around M2, the cervical PDL and gCT demonstrated a localized connective tissue degradation (*), where CD45+ immune cells clustered (arrows). PSR-stained collagen architecture immediately under the ligature lost the thick collagen bundle structure (*). **(C)** Day 3 of ligature placement exhibited localized but increased CD45+ immune cell clustering adjacent to the collagen degradation area (*). **(D)** Day 7 of ligature placement, PDL and gCT tissue degradation increased with inflammatory vascularization (Vas). CD45+ myeloid cells (arrowheads) were observed near the alveolar bone surface and CD45+ lymphocytes (arrows) infiltrated the gCT area. PSR staining lost the typical collagen pattern in gCT and PDL, and a remnant of degraded PDL collagen fiber (white bracket) was attached to the tooth surface. **(E)** Single-cell RNA sequencing (scRNA-seq) *t*-distributed stochastic neighbor embedding (*t*-SNE) projection plots showing the major cell types present within gingival tissue during periodontitis development on Day 0, 1, 4, and 7. Colors indicate cell type, as follows: green, epithelial cells; blue, fibroblasts; pink, endothelial cells; yellow, B cells; red, T cells; and purple, myeloid cells. **(F)** Proportion plots showing the relative amounts of each major cell type on Days 0, 1, 4, and 7.

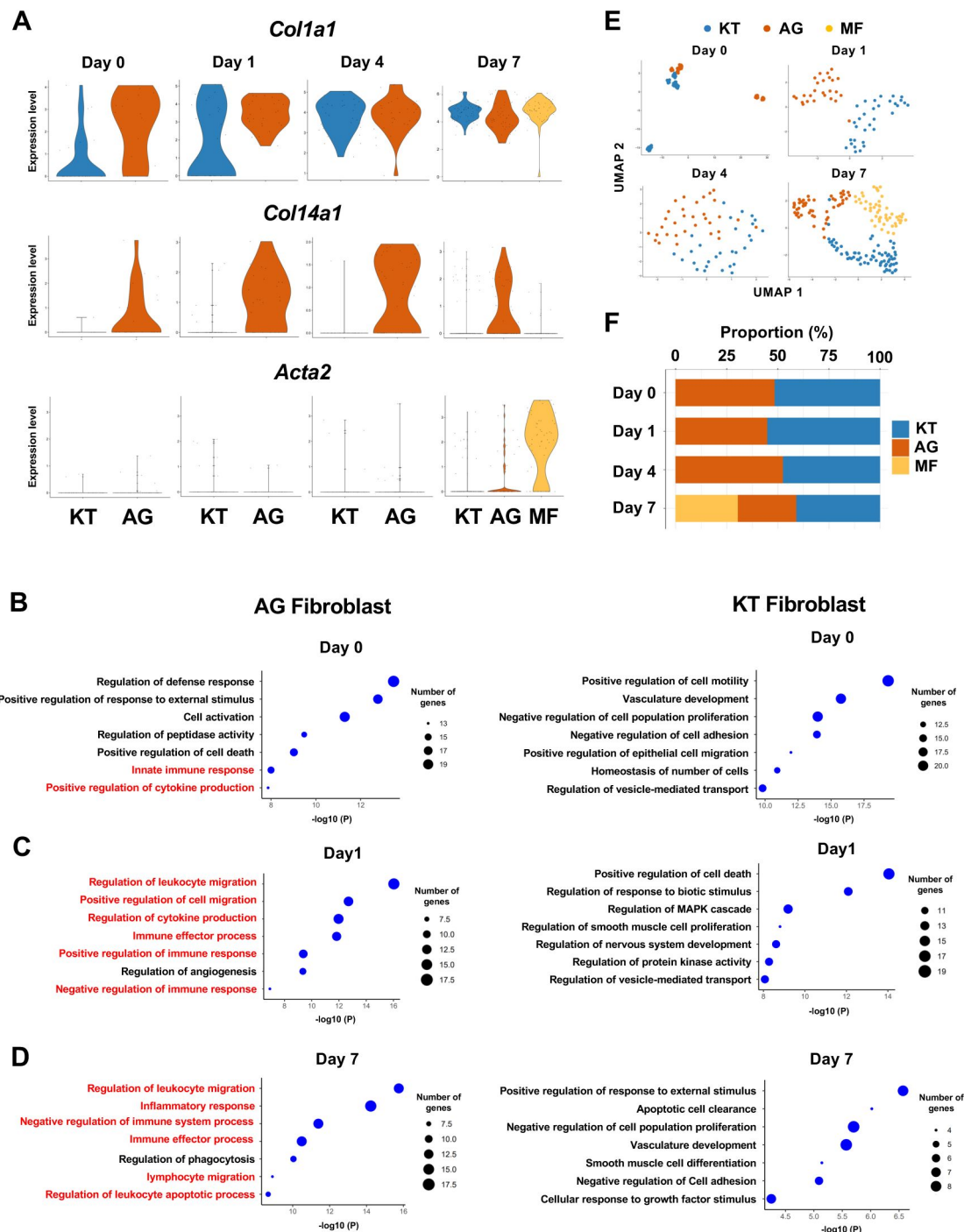


Figure 2.

Fibroblasts activated to guide leukocyte migration (AG fibroblasts) are one of three fibroblast subpopulations in gingival tissue during periodontitis development. (A) Violin plots showing gene expression levels of type I collagen (*Col1a1*), type XIV collagen (*Col14a1*), and smooth muscle aortic actin 2 (*Acta2*) in gingival fibroblast subpopulations during periodontitis development. AG, AG fibroblasts; KT, 'keeping typical phenotype' fibroblasts; MF, myofibroblasts. Gene ontology (GO) enrichment analysis of the biological functions of AG fibroblasts and KT fibroblasts on Day 0 without ligature placement (B) and on Day 1 (C) and Day 7 (D) after ligature placement. Gene clusters related to immune regulation (red) were identified in AG fibroblasts, and these clusters dominate after ligature placement. (E) t-SNE projection plots showing fibroblast subpopulations in gingival tissue during periodontitis development. Colors indicate cell type, as follows: blue, KT fibroblasts; red, AG fibroblasts; and yellow, MFs. (F) Proportion plots showing the relative amounts of each fibroblast subpopulation on Days 0, 1, 4, and 7.

AG fibroblasts and immune surveillance in early periodontitis development

To further characterize AG fibroblasts and their “Regulation of leukocyte migration” phenotype, we analyzed expression of CC motif chemokine ligands (CCLs) and CXC motif chemokine ligands (CXCLs) within fibroblast subpopulations in our scRNA-seq dataset [13]. Results show that AG fibroblasts activated expression of *Ccl8*, *Ccl11*, *Ccl19*, *Cxcl1*, *Cxcl10*, and *Cxcl12* immediately after ligature placement on Days 1 and 4 (Figure 3A). Chemokine expression levels then decreased on Day 7, suggesting that AG fibroblasts might initiate early leukocyte migration into gingival tissue. Given that activation of Toll-like receptors (TLR) is known to increase chemokine expression, we further assessed expression of TLRs in fibroblast subpopulations. We found that AG fibroblasts displayed temporal activation of *Tlr2*, *Tlr3*, and *Tlr4* expression on Days 1 and 4 (Figure 3B). Similarly, the TLR downstream genes *Myd88*, *Irak1*, *Map3k7*, and *RelA* were also expressed on Days 1 and 4 (Figure 3B). These data suggest that AG fibroblasts may guide the establishment of an early inflammatory environment within the gingiva and thereby promote periodontitis pathogenesis.

To validate the presence of AG fibroblasts, Day 1 gingiva and periodontal tissue histological sections were subjected to immunohistochemistry with antibodies against COL14A1 and CXCL12 (Figure 3C). We detected COL14A1- and CXCL12-positive AG fibroblasts localized near gingival epithelial cells in the connective tissue papillae and free gingiva, as well as in the cervical zone of the periodontal ligament (PDL) space (Figure 3C). We note that this localization pattern of AG fibroblasts appears to be highly suitable for early immune surveillance during periodontitis pathogenesis. Thus, we have hypothesized that AG fibroblasts initially sense the pathological stress including oral microbial stimuli through TLRs and secrete inflammatory signals through chemokine expression.

AG fibroblasts induced by maxillary topical application (MTA) of unmethylated cytidine phosphate guanosine oligonucleotide (CpG ODN) and of *Porphyromonas gingivalis* lipopolysaccharide (LPS)

To further characterize the behavior of AG fibroblasts, we used the recently developed maxillary topical application (MTA) model without the placement of a ligature. The MTA model applied oral microbial biofilm directly to the maxillary gingiva and held under a custom-made oral appliance (Figure 4A) for 1 hr. Four days after the topical application, mouse maxillary tissue was harvested for histological analysis and for scRNA-seq (Figure 4B) [8]. We previously reported that oral microbial biofilm and its extracellular polymeric substances such as extracellular DNA, but not planktonic microbes, induced initial periodontal tissue reaction in the MTA model [8]. The present study discretely applied ligands of TLR9 and TLR2/4: unmethylated CpG ODN and *P. gingivalis* LPS, respectively; using the MTA model. CpG ODN induced the expression of cathepsin K (*Ctsk*) in the periodontal ligament (PDL) and gingival connective tissue (Figure 4C) validating the previous report [8]. Furthermore, there were signs of localized *Ctsk*⁺ osteoclastic activities on the surface of alveolar bone (Figure 4C).

The scRNA-seq gingival cell composition of the MTA of CpG ODN and LPS indicated the early periodontitis pattern (Figure 4E) consistent with Day 1 scRNA-seq of the ligature-induced periodontitis (Figure 1F). The MTA of CpG ODN and LPS identified AG fibroblasts along with KT fibroblasts, whereas the MTA of LPS induced the additional MF (Figure 4E). The further analysis demonstrated the expression of CCL and CXCL chemokines by AG fibroblasts activated by the MTA of CpG ODN and LPS (Figure 4F). CpG ODN and LPS are sensed by Toll-Like Receptor 9 (TLR9) and TLR2/4, respectively. The scRNA-seq of MTA of CpG ODN did not capture the transcriptional activation of *Tlr9*, while the downstream effector genes such as *Myd99*, *Irak1* and *RelA* were found upregulated in AG fibroblasts (Figure 4G). The MTA of LPS appeared to upregulate *Tlr2* and *Tlr4*

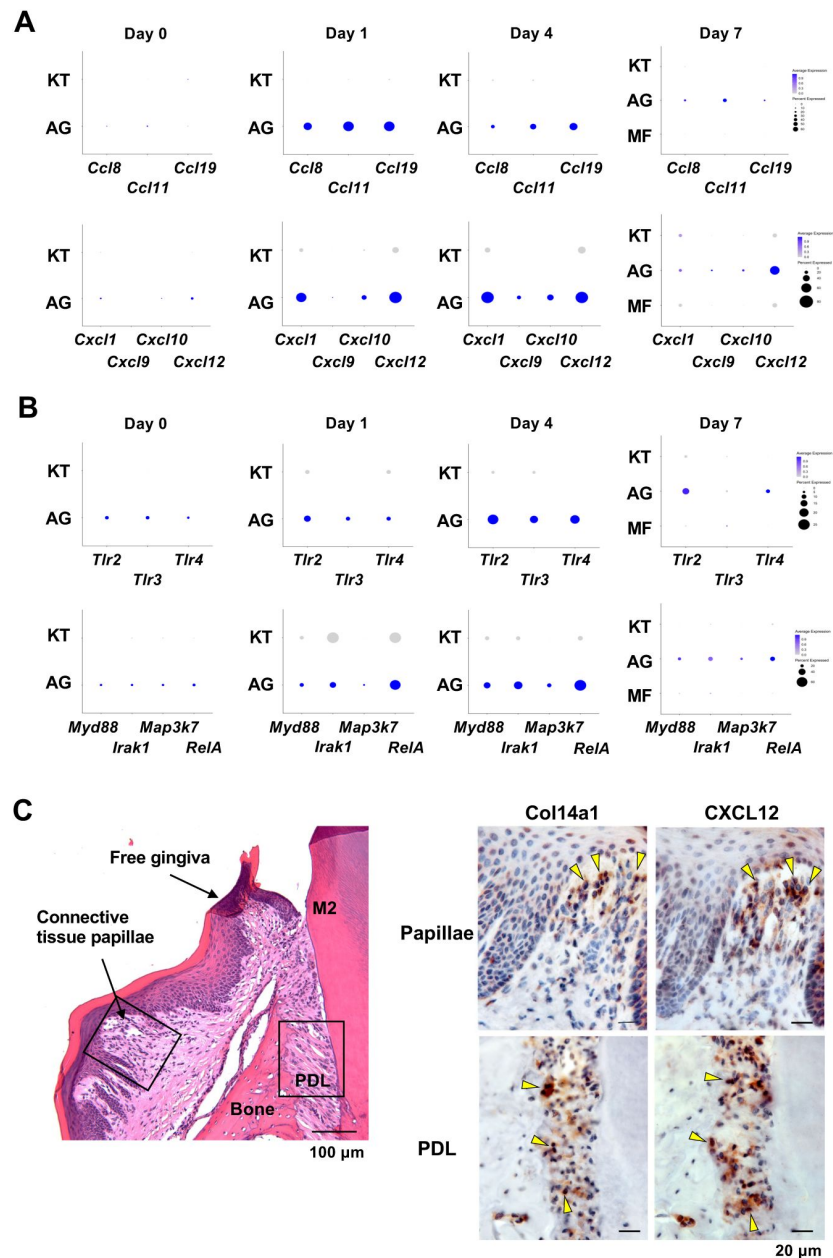


Figure 3.

AG fibroblasts and immune surveillance in periodontitis development. **(A)** Dot plots depicting expression levels of the CCL genes *Ccl8*, *Ccl11*, *Ccl19*, *Cxcl1*, *Cxcl9*, *Cxcl10*, and *Cxcl12* in gingival fibroblast subpopulations during periodontitis development. **(B)** Dot plots depicting expression levels of the Toll-like receptor and related genes *Tlr2*, *Tlr3*, *Tlr4*, *Myd88*, *Irak1*, *Map3k7*, and *Rela* in gingival fibroblast subpopulations during periodontitis development. Upregulation of chemokines and TLR-related molecules is predominantly observed in the AG fibroblast subpopulation. **(C)** Hematoxylin and eosin (HE) staining and immunohistochemical (IHC) staining for COL14A1 and CXCL12 in periodontal tissue on Day 1; scale bars, 100 μ m (HE) and 20 μ m, (IHC). Yellow arrows indicate COL14A1- and CXCL12-positive cells in the connective tissue papillae and periodontal ligament (PDL).

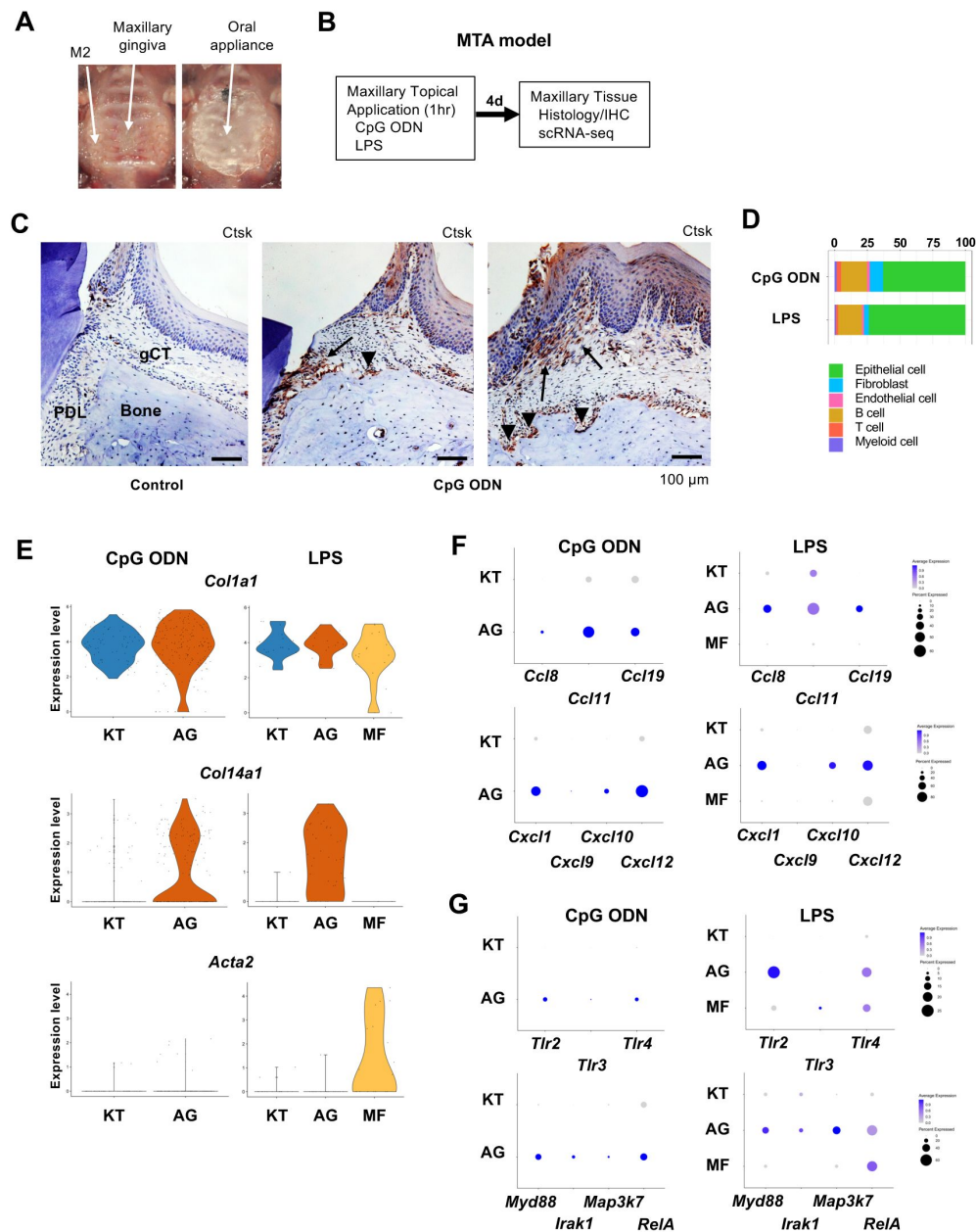


Figure 4.

Characterization of AG fibroblasts in the discrete maxillary topical application (MTA) model using CpG ODN (TLR9 ligand) and LPS (TLR2/4 ligand). **(A)** The MTA model was developed to discretely test the selected oral microbial pathogens. The selected pathogen was topically applied directly on the maxillary gingiva between the molars and covered by a custom fabricated oral appliance for 1 hr. **(B)** In this study, microbial DNA TLR9 ligand, unmethylated CpG ODN and TLR2/4 ligand *P. gingivalis* lipopolysaccharide (LPS) were selected in the MTA model. After 1 hr exposure, mice were returned to the vivarium for 4 days and the maxillary tissue was harvested for histology or scRNA-seq. **(C)** The MTA of CpG ODN developed PDL and gCT degradation evidenced by the expression of cathepsin K (Ctsk; arrows). There were signs of localized bone resorption by Ctsk+ osteoclasts (arrowheads) on the surface of alveolar bone (Bone). **(D)** Gingival cell composition by scRNA-seq of the MTA of CpG ODN or LPS revealed early stage of gingival inflammation, equivalent to Day 1 of the ligature-induced periodontitis model. **(E)** The fibroblastic gene expression signature revealed the presence of KT and AG fibroblasts by the MTA of CpG ODN, whereas the MTA of LPS induced MF. **(F)** AG fibroblasts of the MTA of CpG ODN and LPS were activated to express CCL and CXCL chemokines. **(G)** The MTA of CpG ODN did not upregulate *Tlr9*, whereas the MTA of LPS increased the expression of *Tlr2/4* in AG fibroblasts. However, the both the MTA of CpG ODN and LPS increased the expression of TLR downstream molecules.

gene transcription of AG fibroblasts and of MF albeit at lesser degrees (**Figure 4G**). These results indicated that AG fibroblasts were activated by CpG ODN and LPS through the MTA model, while these microbial elements appeared to exert different reactions.

Myeloid cell composition and activity during periodontitis development

The infiltration of proinflammatory neutrophils into the gingiva has been extensively characterized in this mouse model of periodontitis [11, 38]. Here, we found that macrophages expressing cluster of differentiation 86 (*Cd86*) [39] and integrin subunit alpha X (*Itgax*) [40] were predominant on Day 0 in the myeloid cell fraction from healthy gingiva (**Figure 5A**, **5B**—**figure supplement 3**). In contrast, the proportion of neutrophils expressing CXC motif chemokine receptor (*Cxcr2*) [41] and G0/G1 switch gene 2 (*G0s2*) [42] increased after ligature placement and during periodontitis development from Day 1 to 7 (**Figure 5A**, **5B**—**figure supplement 3**), suggesting continuous neutrophil infiltration into the early and established gingival lesion. Moreover, after ligature placement, gingival neutrophils upregulated the expression of triggering receptor expressed on myeloid cells 1 (*Trem1*), indicating that these cells are activated and participating in the amplification of inflammatory signals [43, 44] (**Figure 5C**). Strikingly, the *Trem1*-expressing activated neutrophils also show upregulation of matrix metalloproteinase 9 (*Mmp9*; **Figure 5C**)—a protein associated with extracellular matrix degeneration within gingival tissue [45]. We further detected expression of tumor necrosis factor (*Tnf*) and transforming growth factor beta 1 (*Tgfb1*) in both macrophages and neutrophils on Day 0 and after ligature placement (**Figure 5D**). Collectively, these myeloid cell behaviors are consistent with those reported in prior studies on periodontitis development, thus validating our scRNA-seq data.

Myeloid cells are also known to stimulate other immune cells through the expression of CCL and CXCL chemokines, many of which are associated with periodontitis development [46]. Our scRNA-seq analysis revealed that macrophages expressed *Ccl2*, *Ccl9*, *Cxcl4*, and *Cxcl6*, and neutrophils expressed *Ccl3*, *Ccl4*, *Ccl6*, *Cxcl2*, and *Cxcl3* throughout periodontitis development (**Figure 5E**, **5F**). These data suggest that chemokines and cytokines produced by macrophages and neutrophils in inflamed tissue may amplify and polarize the immune response toward chronic gingival inflammation.

Role of AG fibroblasts in myeloid cell activation

The interaction between chemokine ligands and their receptors has been extensively studied. Here, to evaluate the interaction between chemokine ligands strongly expressed by AG fibroblasts and chemokine receptors in innate immune cells, we matched AG fibroblast-expressed chemokines to expression of their putative receptors in myeloid cells. Results show that AG fibroblasts may regulate macrophages via the expression *Ccl8* and *Ccl11*, which encode chemokines that can interact with CC chemokine receptors (CCRs) encoded by *Ccr2* and *Ccr5* in macrophages. Similarly, gene expression signatures suggest that AG fibroblast-mediated neutrophil regulation may occur through CCL8–CCR1, CXCL1–CXCR2, and CXCL12–CXCR4 interactions (**Figure 6A**). Notably, all chemokine–receptor pairs are expressed throughout periodontitis development, although expression of factors mediating the interaction between AG fibroblasts and macrophages was decreased on Day 7.

Intercellular communication may occur via other ligand–receptor interactions, which induce downstream target gene expression in recipient cells. Here, we performed NicheNet analysis to identify potential interactions between gingival fibroblasts and myeloid cells, revealing a trend toward increasing ligand–target interactions between fibroblasts and myeloid cells from Day 1 to Day 7 (**Figure 6B**). On Day 1, ligand expression was present primarily in AG fibroblasts, whereas on Day 7, MFs also exhibited prominent ligand expression (**Figure 6C**). Analysis of target gene expression revealed that both macrophages and neutrophils expressed receptors capable of

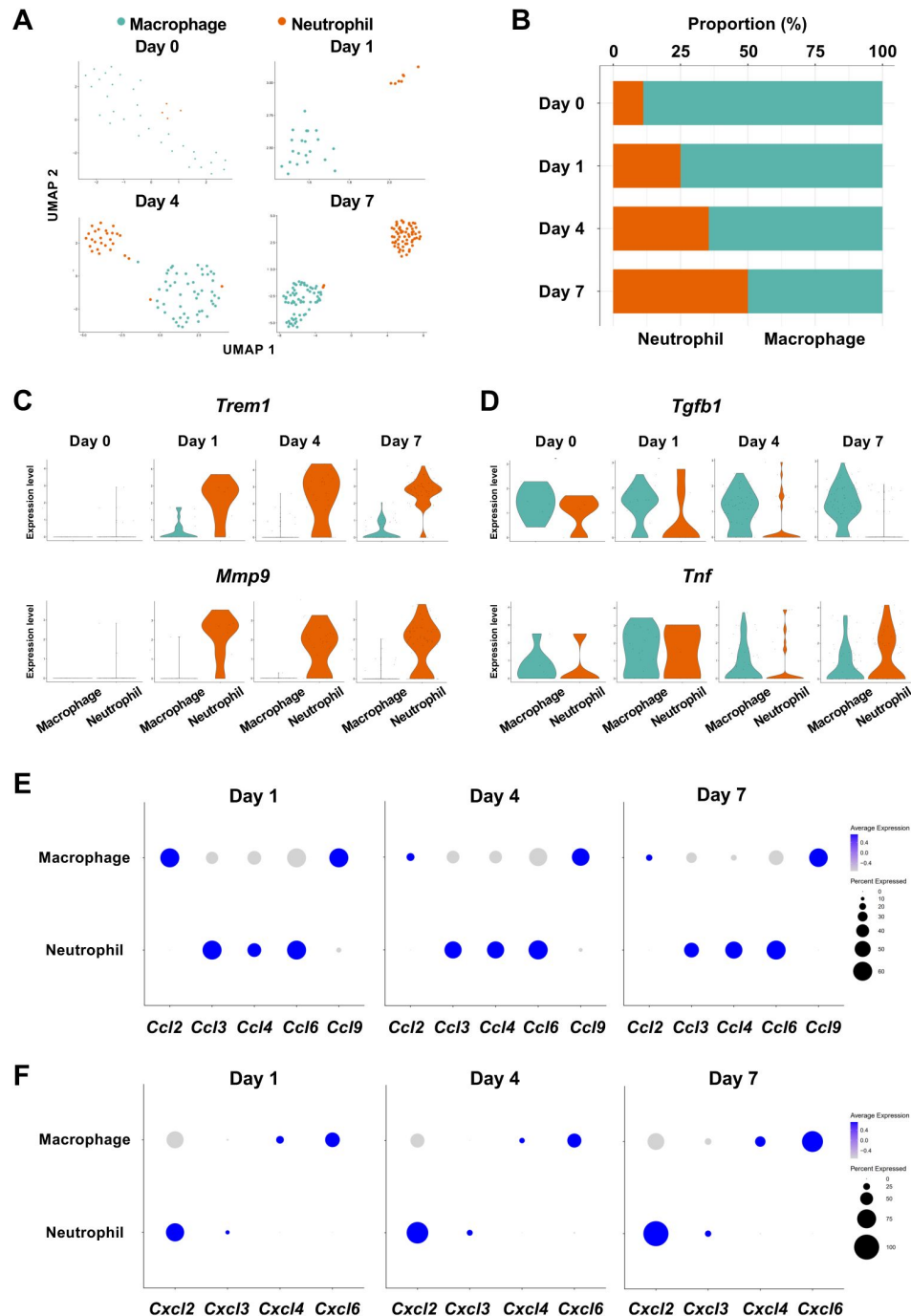


Figure 5.

Myeloid cell composition and activity in gingival tissue during in periodontitis development. **(A)** t-SNE projection plots showing myeloid cell subpopulations in gingival tissue during periodontitis development on Days 0, 1, 4, and 7. Colors indicate cell type, as follows: green, macrophages and red, neutrophils. **(B)** Proportion plots showing the relative amounts of neutrophils and macrophages on Days 0, 1, 4, and 7. **(C)** Violin plots showing *Trem1* and *Mmp9* expression levels in myeloid cells on Days 0, 1, 4, and 7; both genes are upregulated in neutrophils after ligature placement. **(D)** Violin plots showing *Tgfb1* and *Tnf* expression in myeloid cells on Days 0, 1, 4, and 7; no obvious induction is observed in response to ligature placement. **(E)** Dot plots depicting expression levels of the C motif chemokine ligand (CCL) genes *Ccl2*, *Ccl3*, *Ccl4*, *Ccl6*, *Ccl9*. **(F)** The expression of CXC motif chemokine ligand (CXCL) genes *Cxcl2*, *Cxcl3*, *Cxcl4*, and *Cxcl9*. Chemokine expression in myeloid cells was unrelated to progression of gingival inflammation from Day 1 to 7.

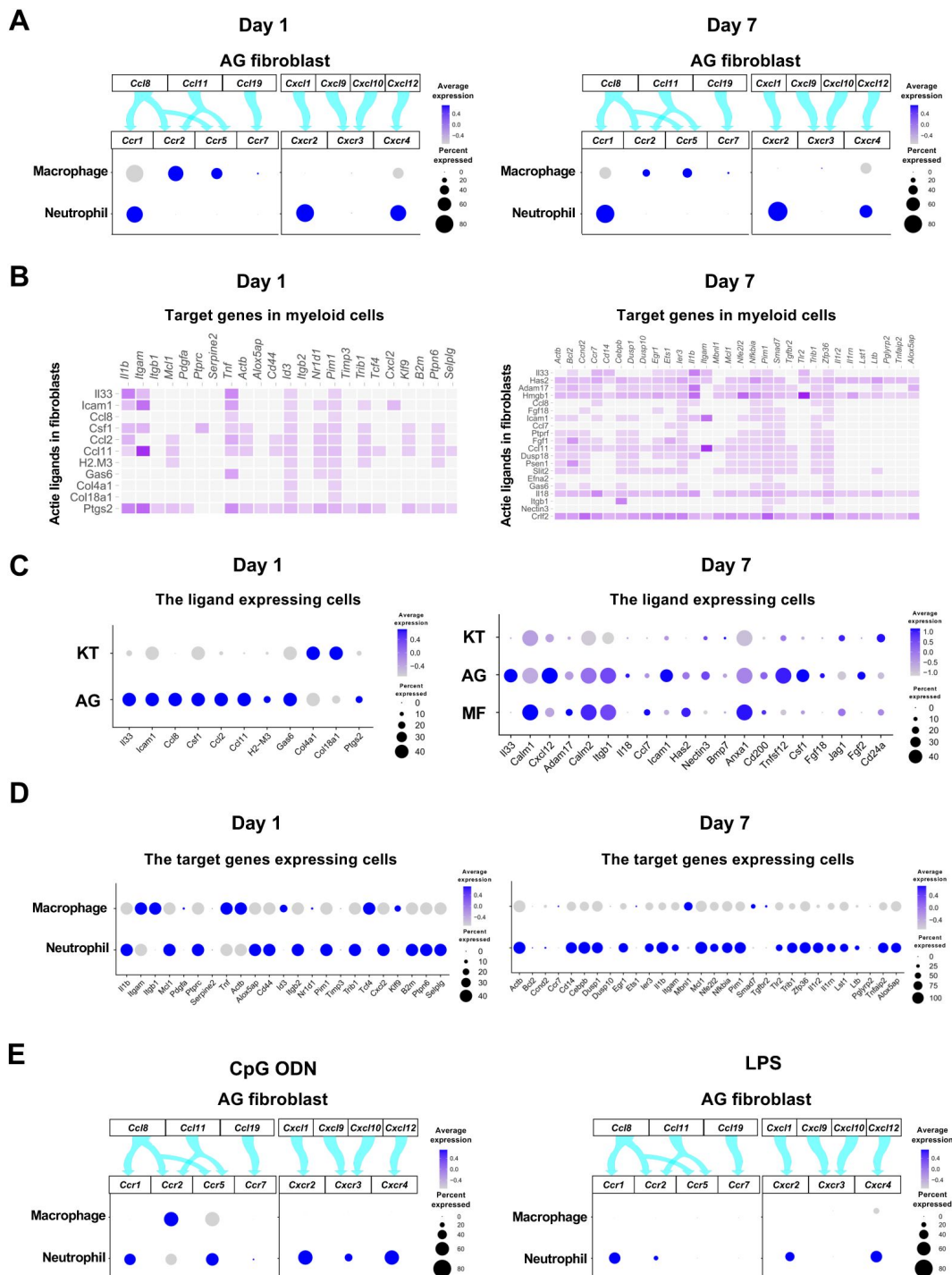


Figure 6.

Role of AG fibroblasts in myeloid cell activation. **(A)** Interaction between chemokine ligands expressed by AG fibroblasts and their putative chemokine receptors expressed by myeloid cells during periodontitis development. Dot plots depicting expression levels of the CC chemokine receptor (CCR) and the CXC chemokine receptor (CXCR) genes *Ccr1*, *Ccr2*, *Ccr5*, *Ccr7*, *Cxcr2*, *Cxcr3*, and *Cxcr4* in myeloid cell subpopulations on Day 1 and Day 7 following ligature placement. **(B)** NicheNet ligand-target matrix indicating the regulatory potential between active ligands expressed in fibroblasts and target genes expressed in myeloid cells from the p-EMT program on Day 1 and Day 7. **(E)** Dot plot depicting expression levels of active ligand genes from panel **(B)** in fibroblast subpopulations on Day 1 and Day 7. **(D)** Dot plot depicting expression levels of target genes from panel **(B)** in myeloid cell subpopulations on Day 1 and Day 7. Results suggest a strong intercellular communication network from AG fibroblasts to neutrophils. **(E)** Interaction between chemokine ligands expressed by AG fibroblasts and their putative chemokine receptors expressed by myeloid cells in the MTA model of CpG ODN and LPS.

interacting with AG fibroblast ligands on Day 1. In contrast, on Day 7, receptor expression was almost exclusively present in neutrophils, suggesting these cells are primarily targeted by MFs and AG fibroblasts at later stages of periodontitis development (**Figure 6D** [↗](#)).

The chemokine-receptor interaction was also suggested between AG fibroblasts and myeloid cells in the MTA of CpG DON (**Figure 6E** [↗](#)). However, in the MTA of LPS, macrophage lacked the detectable expression of chemokine receptor and the AG fibroblastic chemokine interaction appeared to be limited to neutrophils (**Figure 6E** [↗](#)).

Expression of osteoclastogenic cytokines

Macrophage-colony stimulating factor (M-CSF), encoded by *Csf1*, and receptor activator of nuclear factor kappa-B ligand (RANKL), encoded by *Tnfsf11*, are known to play critical roles in osteoclast induction [47 [↗](#)]. We found that in the ligature-induced periodontitis model, the expression of M-CSF was detected in various cell types including fibroblasts (**Figure 7A** [↗](#)), whereas the RANKL expression was more restricted to fibroblasts and T cells (**Figure 7B** [↗](#)). AG fibroblasts consistently expressed M-CSF from Day 0 to Day 7 and neutrophils started to express M-CSF in Day 4 and Day 7 (**Figure 7C** [↗](#)). The expression of RANKL was detected in AG fibroblasts but not in myeloid cells (**Figure 7D** [↗](#)). T cells and type 3 innate lymphoid cells (ILC3; see below) were also shown to express M-CSF and RANKL (**Figure 7C** [↗](#), **7D** [↗](#)).

The MTA of CpG ODN and LPS suggested the AG fibroblasts were one of the predominant cellular sources of M-CSF (**Figure 7E** [↗](#)). By contrast, the expression of RANKL was only detected in AG fibroblasts of the MTA of CpG ODN and LPS did not seem to induce RANKL expression in the MTA model (**Figure 7F** [↗](#)). The osteoclastic activity found in the MTA of CpG ODN (**Figure 4D** [↗](#)) might be induced by M-CSF and RANKL derived from AG fibroblasts.

Helper T (Th) cells, cytotoxic T (Tc) cells, regulatory T (Treg) cells and innate lymphoid cells (ILCs) in periodontitis development

Our scRNA-seq dataset found that the Day 0 healthy gingiva exclusively contained Th cells expressing *Cd4* [48 [↗](#)]. However, after ligature placement, Tc cells expressing *Cd8* [48 [↗](#)] emerged, and on Day 7, *Cd4⁺Zbtb16⁺* ILCs and *Cd4⁺Foxp3⁺* Treg cells [49 [↗](#)] were also detected (**Figure 8A** [↗](#), **8B** [↗](#)). The ILCs expressed *Nfil3*, a basic leucine zipper (bZIP) transcription factor required for ILC development [50 [↗](#)] (**Figure 8C** [↗](#)). However, to our surprise, expression of *Rorc*, *Il17a*, and *Il17f* was detected in ILCs (**Figure 8C** [↗](#)), indicating that this gingival subpopulation is predominantly composed of type 3 ILCs (ILC3s). ILC3s and Th17 cells share similar regulatory functions. However, the role of ILC3s in the development of periodontitis has not been fully deciphered.

ILC3s are critical for cervical alveolar bone resorption in the mouse periodontitis model

We thus examined the role of ILC3s in periodontitis pathogenesis by measuring ligature-induced gingival defects and alveolar bone resorption in *Rag2^{-/-}* mice, which lack functional B, Th, and Tc cells, and *Rag2*-IL-2 receptor common gamma (*Il2rg*) double-knockout mice (*Rag2^{-/-}Il2rg^{-/-}*), lacking all lymphocytes including ILCs. After ligature placement, we found that alveolar bone loss was decreased in *Rag2^{-/-}* mice and nearly eliminated in *Rag2^{-/-}Il2rg^{-/-}* mice (**Figure 8D** [↗](#)). MicroCT image analysis indicated a better preservation of alveolar bone structure in *Rag2^{-/-}Il2rg^{-/-}* mice, relative to the other groups (**Figure 8E** [↗](#)—figure supplement 4A [↗](#), 4B [↗](#)). However, gingival defects developed similarly in wild-type (WT), *Rag2^{-/-}*, and *Rag2^{-/-}Il2rg^{-/-}* mice (figure supplement 4C [↗](#), 4D [↗](#)).

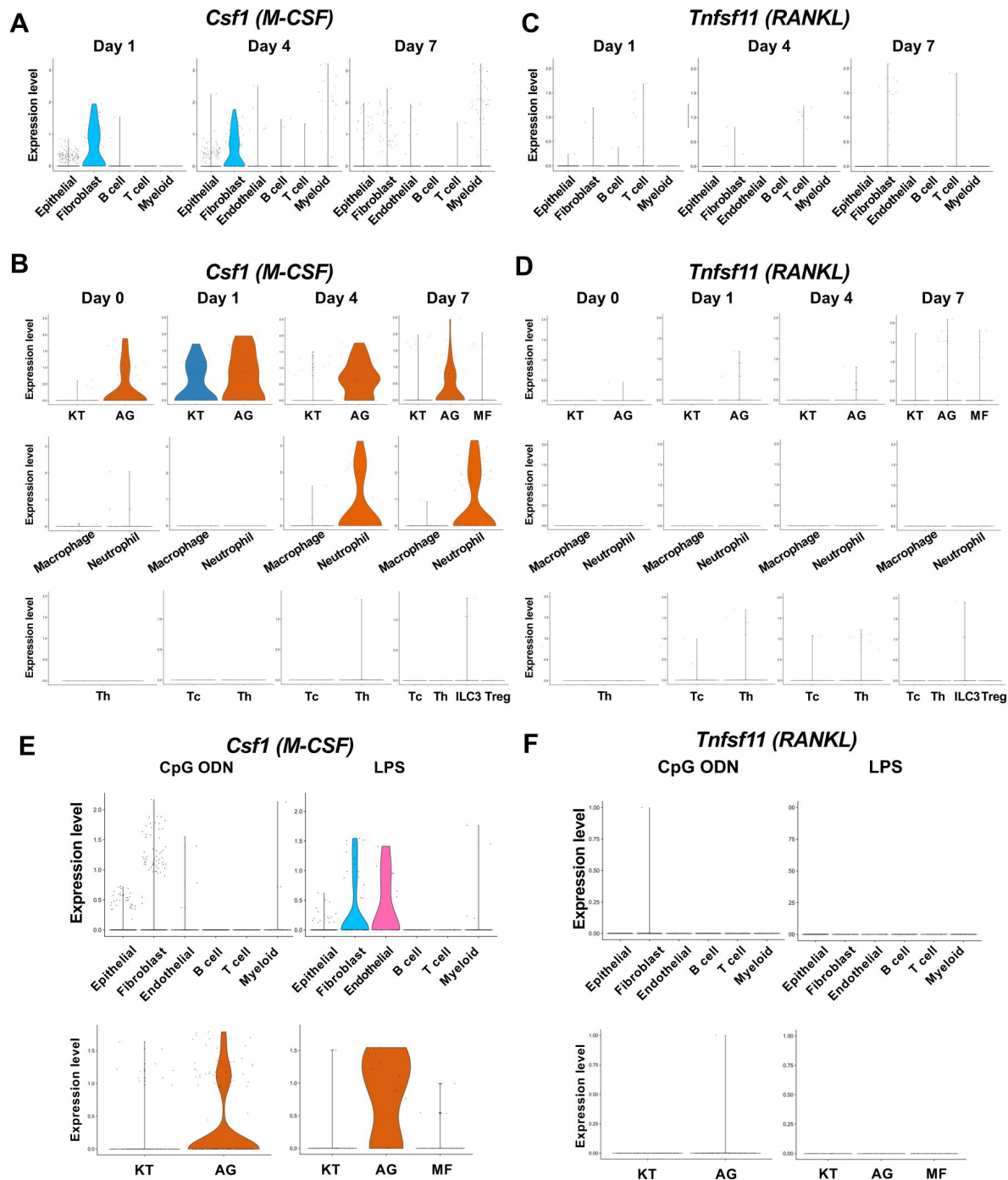


Figure 7.

The cellular source of osteoclastogenic cytokines: M-CSF and RANKL in the ligature-induced model and the MTA model. **(A)** Violin plots showing expression levels of the macrophage-colony stimulating factor (M-CSF)-encoding gene, *Csf1* (M-CSF), in each major cell type from the ligature-induced periodontitis model. **(B)** Violin plots showing expression levels of *Csf1* in fibroblast subpopulations, myeloid cell subpopulations, and T cell subpopulations. **(C)** Violin plots showing expression levels of the nuclear factor kappa-B ligand (RANKL)-encoding gene, *Tnfsf11*, in each major cell type. **(D)** Violin plots showing expression levels of *Tnfsf11* in fibroblast subpopulations, myeloid cell subpopulations, and T cell subpopulations. **(E)** Violin plots of *Csf1* expressing cells in the MTA model. **(F)** Violin plots of *Tnfsf11* expressing cells in the MTA model. AG fibroblasts predominantly expressed *Tnfsf11*.

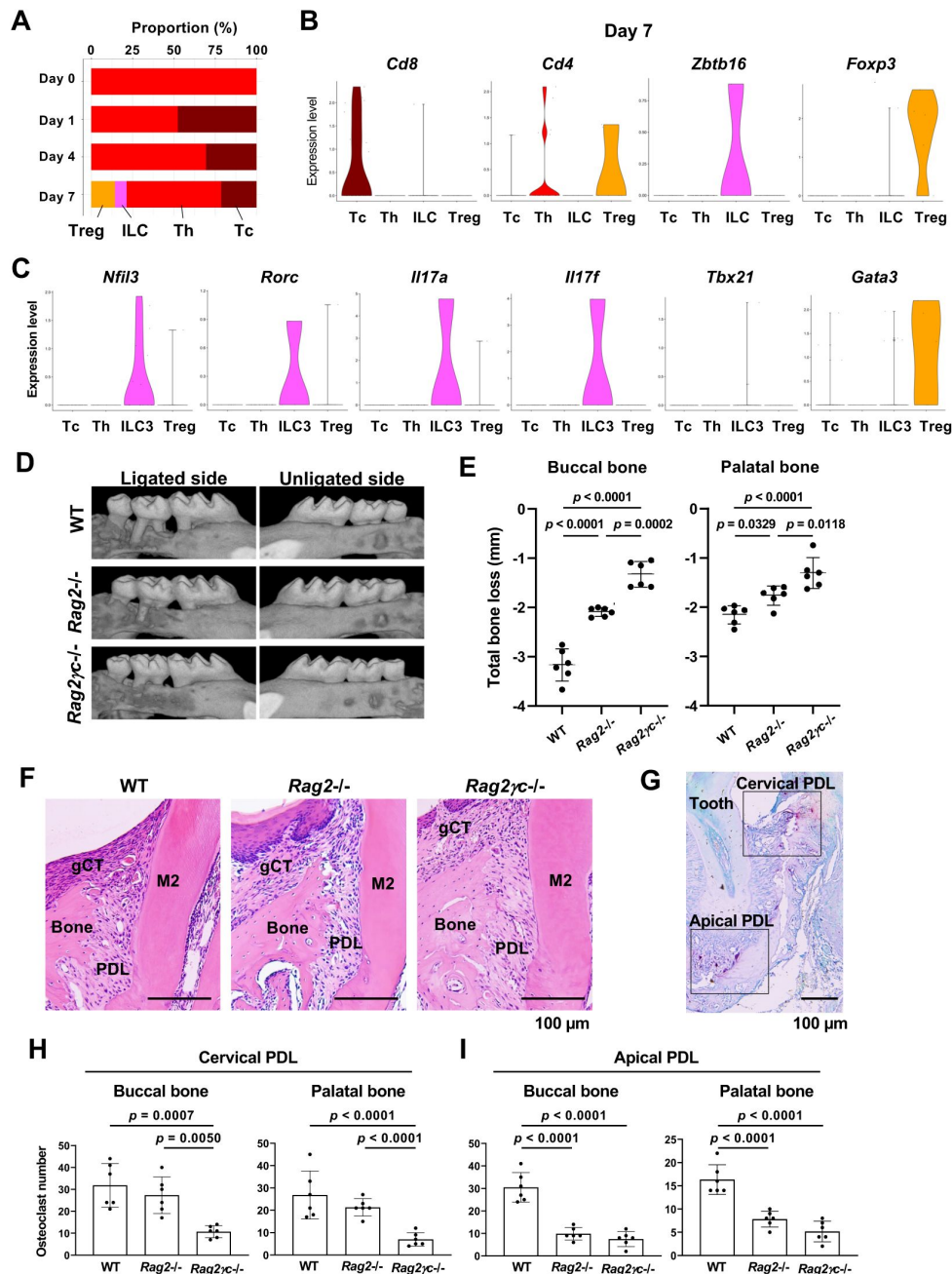


Figure 8.

ILC3s are critical for cervical alveolar bone resorption in the ligature-induced periodontitis development. **(A)** Proportion plots showing the relative amounts of T cell subpopulations in gingival tissue during periodontitis development. Treg, T regulatory cells; ILC, innate lymphoid cells; Th, T helper cells; Tc, cytotoxic T cells. **(B)** Violin plots showing expression levels of the T cell marker genes *Cd8* (Tc), *Cd4* (Th), *Zbtb16* (ILC), and *Foxp3* (Treg) on Day 7 following ligature placement. **(C)** Violin plots showing expression levels of *Nfil3*, *Rorc*, *Il17a*, *Il17f*, *Tbx21*, and *Gata3* on Day 7 following ligature placement. These gene signatures indicate that gingival ILCs primarily comprise type 3 ILCs (ILC3s). **(D)** Representative microCT images of the maxilla taken from the lateral view for the ligated side and from the contralateral view for the unligated side. **(E)** Alveolar bone loss was determined from the total distance between the CEJ and the ABC of the buccal bone or palatal bone at six sites in the ligated side ($n = 6$). **(F)** HE staining of the periodontal tissue on Day 7. gCT, gingival connective tissue; Bone, alveolar bone; PDL, periodontal ligament; scale bars, 100 μm . **(G)** Tartrate-resistant acid phosphatase (TRAP) staining of periodontal tissue from WT mice on Day 7; scale bar, 100 μm . Total number of TRAP-positive cells in a 0.01 mm^2 -area of the buccal and palatal bone in the cervical PDL site **(H)** and apical PDL site **(H)** ($n = 6$). Significance was determined by ANOVA, with Tukey's multiple-comparison test. Data are presented as mean values $\pm$ SD; $p < 0.05$ was considered significant.

Histologically, osteoclastic resorption lacunae were observed on the alveolar bone surface at the cervical PDL and tooth apex PDL zones in WT mice (**Figure 8F** [↗](#), **8G** [↗](#)). In addition, we detected a significant decrease in the number of tartrate-resistant acid phosphatase (TRAP)-positive osteoclasts in the cervical PDL zone of *Rag2^{yc-/-}* mice (**Figure 8H** [↗](#)) and in the apical PDL zone of both *Rag2^{-/-}* and *Rag2^{yc-/-}* mice (**Figure 8I** [↗](#)). Collectively, these data suggest that ILC3s, not Th17 cells, are responsible for cervical alveolar bone resorption in the mouse periodontitis model, a pathological phenotype consistent with human disease (**figure supplement 4E** [↗](#), **4F** [↗](#)).

The role of AG fibroblasts and neutrophils in ILC3 development in periodontitis

Based on our present data, we hypothesize that ILC3s within the gingival tissue play a pathological role in cervical alveolar bone resorption in our mouse model. We therefore aimed to identify the cells that promote ILC3 development in mice. Similar to Th17 cells, ILC3 development was shown to be triggered by IL-6 and IL-23a [[51](#) [↗](#), [52](#) [↗](#)]. A survey of our scRNA-seq data identified epithelial cells, fibroblasts, and myeloid cells as the source of *Il6* in gingival tissue (**Figure 9A** [↗](#)). On Days 1 and 4 following ligature placement, AG fibroblasts primarily expressed *Il6*, whereas neutrophils became the predominant source on Day 7. We further found that gingival epithelial cells comprise at least four different subpopulations, plus an additional group displaying an epithelial–mesenchymal transition (EMT) phenotype on Day 7; *Il6* was expressed by several of these epithelial subpopulations, including the EMT subgroup (**Figure 9A** [↗](#)—**figure supplement 5A–D** [↗](#)). *Il23a* was also detected AG fibroblasts and myeloid cells (**Figure 9B** [↗](#)), with expression present in all epithelial cell subsets at various points in periodontitis development (**Figure 9B** [↗](#)—**figure supplement 5D** [↗](#)).

Lastly, we evaluated potential interactions between chemokine ligands expressed by AG fibroblasts and neutrophils and chemokine receptors in innate immune cells. Our data suggest the presence of a chemokine–receptor association between ILC3s and both AG fibroblasts (**Figure 9C** [↗](#)) and neutrophils (**Figure 9D** [↗](#)), although interactions with other innate immune cells are also possible. NicheNet analysis further identified potential ligand–target gene associations between lymphocytes, including ILC3s, and both fibroblasts (**Figure 9E** [↗](#)) and myeloid cells (**Figure 9F** [↗](#)). Ligand expression was more prominent in AG fibroblasts than in other fibroblast subpopulations (**Figure 9G** [↗](#)) and elevated in neutrophils relative to macrophages (**Figure 9H** [↗](#)). Additionally, target gene expression was detected in ILC3s, although it was not specific to these cells (**Figure 9I** [↗](#), **9J** [↗](#)). Thus, in total, our data suggest a regulatory role for a newly identified AG fibroblast subpopulation in the gingiva, which appears to orchestrate chronic gingival inflammation, at least in the early stages, and to promote alveolar bone resorption via stimulation of neutrophils and ILC3s.

Discussion

The present study revealed that the gene signature of a unique and previously uncharacterized subpopulation of gingival fibroblasts, referred to as AG fibroblasts, could possess the functional capability to serve as an oral immune surveillant and to orchestrate the initiate gingival inflammation. Oral barrier immunity presents a complex interaction between different types of immune cells to protect and maintain the oral environment and structure. Given the frequent exposure to a diverse commensal and pathological microbiome, physical and chemical insults, and dietary and airborne antigens, the oral barrier immune mechanism must resiliently establish the highly tolerant homeostasis [[53](#) [↗](#)]. This oral immune homeostasis mechanism is not yet fully understood to date; however, the aberrant oral immune response and gingival chronic inflammation leading to periodontitis have provided an important clue to elucidate the oral barrier immunity.

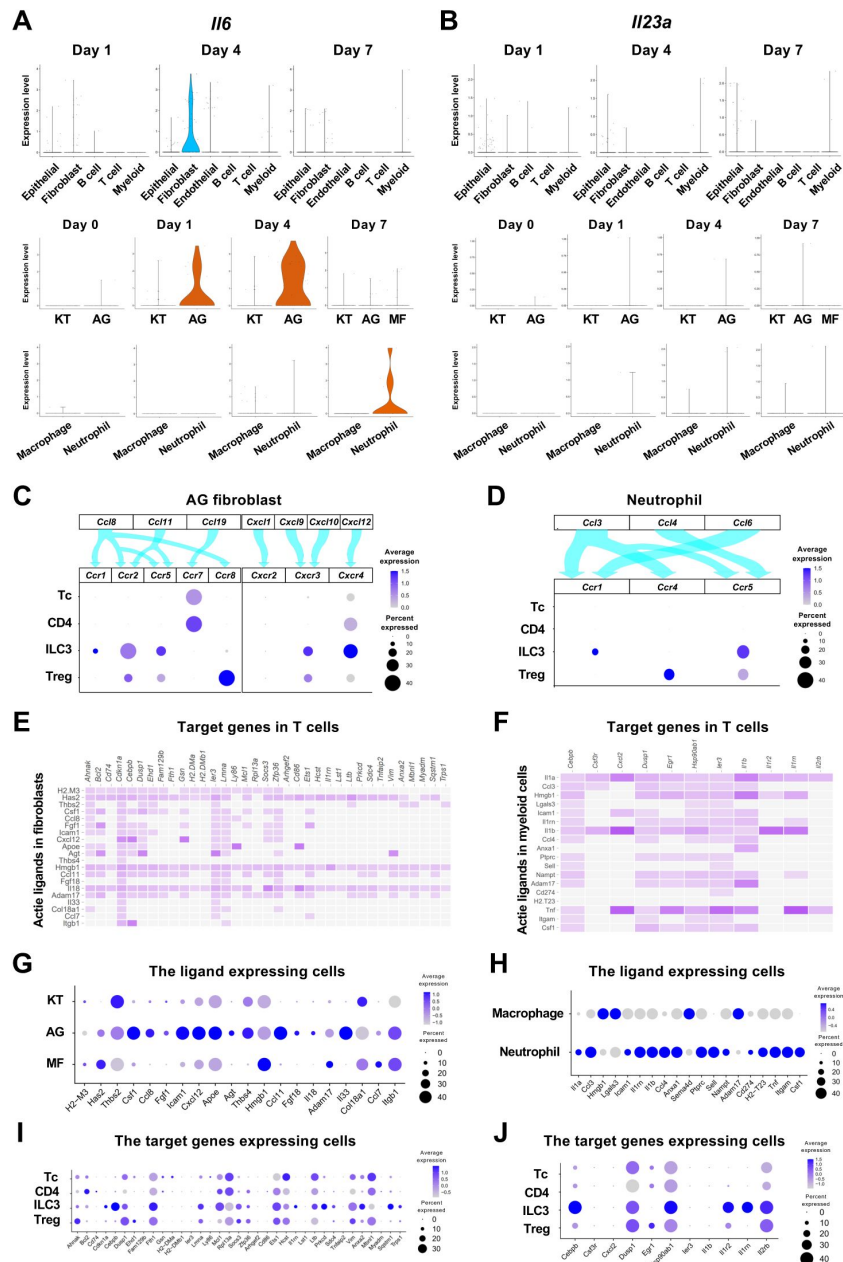


Figure 9.

The role of AG fibroblasts and neutrophils in ILC3 development in periodontitis. Violin plots showing expression levels of the genes encoding interleukin (IL)-6 (*Il6*) (**A**) and IL-23 (*Il23a*) (**B**) in each major cell type, fibroblast subpopulations, and myeloid cell subpopulations during periodontitis development. (**C**) Interaction between chemokine ligands strongly expressed by AG fibroblasts and their putative chemokine receptors expressed by T cells, including ILCs. Dot plots depict gene expression levels of *Ccr1*, *Ccr2*, *Ccr5*, *Ccr7*, *Ccr8*, *Cxcr2*, *Cxcr3*, and *Cxcr4* in T cell subpopulations on Day 7 following ligature placement. (**D**) Interaction between chemokine ligands strongly expressed by neutrophils and their putative chemokine receptors expressed by T cells. Dot plots depicting gene expression levels of *Ccr1*, *Ccr4*, and *Ccr5* in T cell subpopulations on Day 7 following ligature placement. (**E**) NicheNet ligand-target matrix denoting the regulatory potential between active ligands in fibroblasts and target genes in T cells from the p-EMT program on Day 7 following ligature placement. (**F**) NicheNet ligand-target matrix denoting the regulatory potential between active ligands in myeloid cells and target genes in T cells from the p-EMT program on Day 7 following ligature placement. (**G**) Dot plot depicting expression levels of active ligand genes from panel (**E**) in fibroblast subpopulations. (**H**) Dot plot depicting expression levels of active ligand genes from panel (**F**) in myeloid cell subpopulations. (**I**) Dot plot depicting expression levels of target genes from panel (**E**) in T cell subpopulations. (**J**) Dot plot depicting expression levels of target genes from panel (**F**) in T cell subpopulations.

The previous studies suggested gingival intraepithelial γ 8T cells [54], a subset of neutrophils [55], macrophages [56], and dendritic cells [57] should serve the candidate immune surveillant in the oral barrier tissue. These immune cells recognize the pathological signals through the TLR sensing mechanism. $TLR2^{-/-}$, $TLR4^{-/-}$, $TLR2\&4^{-/-}$ [58] and $TLR9^{-/-}$ [59, 60] mice exhibited the reduced alveolar bone loss in various mouse periodontitis models, suggesting that TLRs indeed played an important role in developing discordant oral barrier immunity. However, these studies did not identify the TLR-carrying surveillant cells centrally involved in initiating the chronic gingival inflammation.

It has been suggested that TLR-expressing gingival fibroblasts may regulate innate immune responses [61, 62]. In the present study, we found that not all gingival fibroblasts acquired an immune-sensing capability, but rather, AG fibroblasts represented a distinct fibroblast subpopulation, capable of responding to microbial and tissue damage signals to serve initiate immune surveillance. To test if TLR ligands stimulate the AG fibroblast activation, we applied a newly developed mouse system employing the discrete MTA model [8]. The present study used TLR9 ligand: CpG ODN; and TLR2/4 ligand: LPS. Topical application of these TLR ligands to the maxillary gingival tissue activated AG fibroblasts and increased expression of CC and CXC chemokines (**Figure 4F**). It was noted that the CpG ODN application did not upregulate *Tlr9* expression, while the LPS application increased *Tlr2/4* expression. *TLR9* mRNA level has not been fully correlated to the chronic inflammatory diseases; however, the pathological activation of TLR9 caused the discordant downstream inflammation [63]. In the present study, the differential expression of the TLR downstream signaling molecules suggested the ligand specific response by AG fibroblasts. In fact, both stimulants increased the expression of myeloid differentiation marker 88 (*Myd88*), IL1R-associated kinase (*Irak*), mitogen-activated protein kinase (*Map3k7*) and NF- κ B subunit *RelA* (**Figure 4G**), suggesting that the corresponding TLRs were indeed stimulated [64, 65]. The *Tlr* expression pattern of Day 1 AG fibroblasts was similar to that of the MTA of CpG ODN, whereas the Day 4 AG fibroblasts resembled the *Tlr* expression pattern of the MTA of LPS (**Figure 3B**). It is tempting to speculate that microbial DNA/TLR9 activation of AG fibroblasts may initiate the early pathological process followed by the LPS/TRL2/4 stimulation in the more established periodontal inflammation. Taken together, we hypothesize that AG fibroblasts sensitively detect the microbial signals and play a previously unrecognized role in innate immune regulation during the early periodontitis development.

One of the major observations in the present study was that AG fibroblasts clearly upregulated the chemokine expression only after the pathological stimulation. CCL and CXCL chemokines orchestrate the chronic inflammation through the ligation and activation of their putative chemokine receptors in a specific mechanism. For example, neutrophils express a relatively limited number of chemokine receptors—CXCR2, CXCR4, and CCR1 [66, 67]. Therefore, neutrophil migration and trafficking to the affected gingiva as well as its pathological transformation in chronic periodontitis patients [11, 68, 69] may be guided through the specific chemokine-receptor interaction. Consistent with prior studies, we detected the expression of these chemokine receptors in gingival neutrophils, and they were linked to the chemokine ligands expressed by AG fibroblasts—CXCL1, CXCL12, and CCL8, respectively (**Figure 6**). In addition, neutrophils can respond to various chemo-attractants that modulate and finetune various cellular behaviors, such as migration direction, adhesion strength, and functional heterogeneity [70]. Here, we observed evidence for substantial intracellular communication between AG fibroblasts and neutrophils through ligand–target gene interactions (**Figure 6**). Although validation of each putative ligand–receptor interaction is beyond the scope of this study, our data suggest that AG fibroblasts support neutrophil trafficking in both early and established periodontitis lesions.

A primary pathological consequence of periodontitis is uncontrolled alveolar bone resorption, leading to tooth loss, which is thought to be mediated by osteoimmune pathways involving IL-17–expressing Th17 cells [71]. Surprisingly, analysis of our scRNA-seq revealed that the expression

of *Rorc*, *Il17a*, and *Il17f* was not detected in CD4⁺ Th cells, but rather, in lymphocytes with ILC characteristics (**Figure 8**). ILCs share phenotypic and functional features with CD4⁺ T cells, although they lack antigen-specific T cell receptors [50]. The gingival ILCs expressed *Rorc*, *Il17a*, and *Il17f*, but not *Tbx21* and *Gata3*, indicating that they were type 3 ILCs (ILC3s). Notably, recent clinical studies have reported the presence of ILCs in gingiva from human periodontitis patients [72, 73] and leptin receptor-deficient mice [74], although their pathological contributions have not been fully elucidated.

In the present study, we further explored the role of ILC3s in periodontitis-associated bone loss using *Rag2*^{-/-} and *Rag2*^{yc}^{-/-} mice. *Rag2*^{-/-} mutation prevents V(D)J recombination required for generating immunoglobulin and T cell receptors, resulting in the production of functionally immature B and T cells, including Th17 cells. However, ILCs do not undergo genomic receptor rearrangements and, thus, are unaffected by *Rag2*^{-/-} mutation. In contrast, *Rag2*^{yc}^{-/-} mice have the additional *Il2rg*^{-/-} mutation, which disables common γ chain cytokines (γ c). Therefore, in addition to non-functional B and T cells, these animals also have defective γ c-dependent ILCs. The ligature-placed *Rag2*^{-/-} mice exhibited a reduced number of osteoclasts in the apical PDL area, indicative of reduced bone resorption in this root apical region. By contrast, *Rag2*^{yc}^{-/-} mice demonstrated a significant loss of bone resorption in both the cervical and apical PDL areas (**Figure 8**). Human periodontitis induces the localized alveolar bone resorption at the cervical PDL zone interfacing the gingival inflammatory lesion. In contrast, apical alveolar bone resorption is observed in clinical cases of root canal infection (apical periodontitis). Therefore, our findings suggest that ILC3s, which are differentially impacted by *Rag2*^{yc}^{-/-} and *Rag2*^{-/-} mutations, may be primarily responsible for human periodontitis-like cervical alveolar bone resorption near the site of gingival inflammation.

Differentiation of ILC3s and Th17 cells is mediated by similar environmental cues, with IL-6 and IL-23a playing critical roles in both pathways [75, 76]. Our present scRNA-seq data (**Figure 9**) suggest that AG fibroblasts and neutrophils are the primary cellular sources of *Il6* during the early and later stages, respectively, of periodontitis development. In contrast, *Il23a* was found to be expressed by a variety of cell types, such as AG fibroblasts, myeloid cells, and multiple subsets of epithelial cells, including those with an EMT phenotype [77]. Further, CC and CXC chemokine–receptor associations between ILC3s and both AG fibroblasts and neutrophils appeared to be non-specific. Therefore, our data suggest that rather than a specific trigger, the collective gingival environment, which includes AG fibroblasts, might contribute to ILC3 differentiation.

In conclusion, based on our present findings, we propose that a previously unrecognized AG fibroblast subpopulation in the gingiva can facilitate immune surveillance and participate in the pathological regulation of innate immune cells, such as proinflammatory neutrophils, within oral barrier tissue (**Figure 10**). Moreover, we hypothesize that ILC3s in the inflamed gingiva play a critical role in pathological alveolar bone resorption in the mouse model of periodontitis and, potentially, in human disease. Thus, the newly proposed AG fibroblast–neutrophil–ILC3 axis may hold valuable clues for unraveling the pathological mechanisms underlying periodontitis development. Moreover, these findings also provide a basis for investigation of new preventive and therapeutic strategies to contain oral barrier inflammation and potentially sever the link between periodontitis and debilitating non-communicative metabolic and cardiovascular diseases.

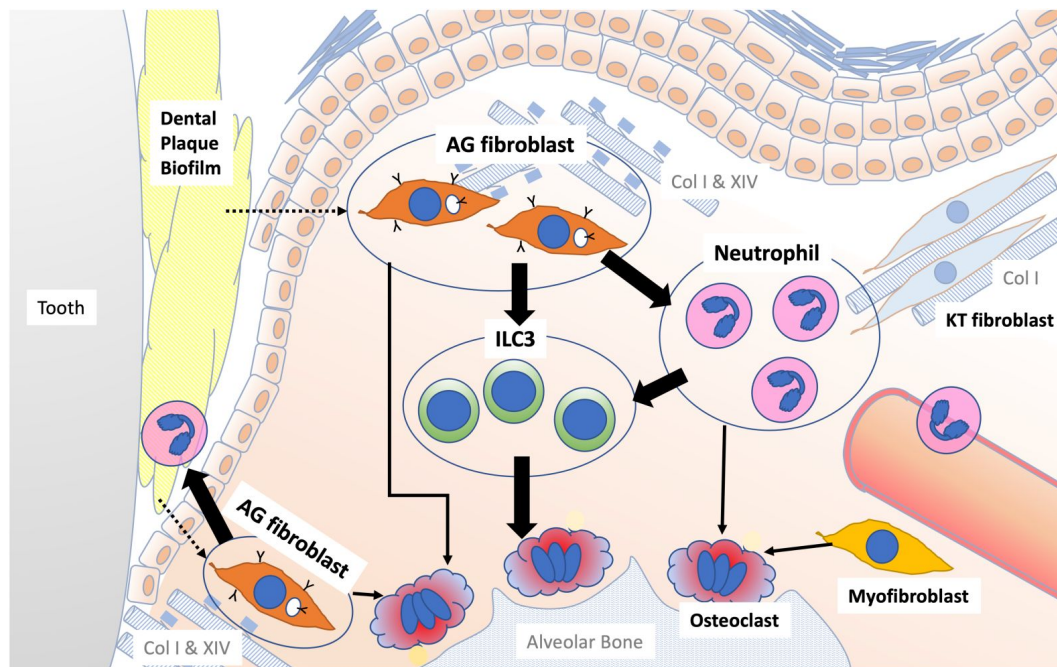


Figure 10.

Schematic overview of the newly proposed AG fibroblast–neutrophil–ILC3 axis. We propose that periodontal inflammation is initiated by the activation of AG fibroblasts, which secrete chemokines and cytokines that recruit neutrophils to sites of tissue damage. Activated neutrophils and AG fibroblasts, in turn, activate ILC3s, leading to production of proinflammatory IL-17 cytokines. Ultimately, cervical alveolar bone resorption is facilitated by a localized osteoclastogenic environment, induced by activated ILC3s, together with AG fibroblasts, neutrophils, myofibroblasts, and gingival epithelial cells, including those with an epithelial–mesenchymal transition (EMT) phenotype.

Animal care

All protocols for animal experiments were reviewed and approved by the University of California Los Angeles (UCLA) Animal Research Committee (ARC# 2003-009) and followed the Public Health Service Policy for the Humane Care and Use of Laboratory Animals and the UCLA Animal Care and Use Training Manual guidelines. C57BL/6J WT, *Rag2*^{-/-}, and *Rag2γc*^{-/-} mice were purchased from the Jackson Laboratory (Bar Harbor, ME, USA). Animals had free access to regular rodent diet and water and were maintained in standard housing conditions with 12-h light/dark cycles in the Division of Laboratory Animal Medicine at UCLA.

Evaluation of gingival defect and alveolar bone resorption in a ligature-induced mouse model of periodontitis

A silk thread was gently tied around the left maxillary second molar of 8- to 12-week-old female WT, *Rag2*^{-/-}, and *Rag2γc*^{-/-} mice under general inhalation anesthesia with isoflurane (Henry Schein, Melville, NY, USA). On Days 1, 3, and 7 after ligature placement, WT mice were randomly chosen and euthanized by 100% CO₂ inhalation. WT mice without ligature placement were used as Day 0 pre-periodontitis control. *Rag2*^{-/-} and *Rag2γc*^{-/-} mice were euthanized on Day 7. The palatal tissue was digitally photographed, and maxillae were harvested and fixed in 10% buffered formalin (Thermo Fisher Scientific, Waltham, MA, USA). The gingival defect area was measured from digital photographs, using the ImageJ Java-based image-processing program (NIH, Bethesda, MD, USA), and normalized to the circumferential area of the maxillary first molar. Fixed maxillae were subjected to microCT imaging at an energy level of 60 kV and 166 μA, and 3D images were reconstructed from microCT scans (Skyscan 1275: Bruker, Billerica, MA, USA). Alveolar bone loss was assessed at three sites (mesiobuccal cusp, distobuccal cusp, and distal cusp) of the first molar, two sites (mesiobuccal cusp and distobuccal cusp) of the second molar, and one site (buccal cusp) of the third molar by measuring the distance from the cementoenamel junction (CEJ) to the alveolar bone crest (ABC) on the buccal and palatal side of the alveolar bone. Total bone loss was calculated from the six-site total CEJ–ABC distance. The bone volume/total volume (BV/TV) ratio, bone surface, trabecular number (Tb.N), and trabecular thickness (Tb.Th) in the buccal side of alveolar bone of the second molar were determined using the proprietary analysis program (CTan: Bruker, Billerica, MA, USA). Statistical analysis to assess differences among multiple experimental groups was performed using one-way analysis of variance (ANOVA) with Tukey's multiple-comparison test; *p* < 0.05 was considered to be statistically significant.

Evaluation of gingival effect in the maxillary topical application (MTA) model

We have developed a method to apply chemical therapeutic agents topically to the mouse maxillary tissue [24, 78]. We used the maxillary topical application (MTA) model to apply oral microbial components [8]. The present study used the MTA model using 1 μg/ml unmethylated CpG oligonucleotide (CpG ODN: InvivoGen, San Diego, CA) or *P. gingivalis* lipopolysaccharide (LPS: InvivoGen). First, a custom-made oral appliance was fabricated using clear dental resin, covering the maxillary/palatal tissue between the molars. Mice were anesthetized and placed on a supine position. CpG ODN (3 μl) or LPS (3 μl) was pipetted over the maxillary tissue and the oral appliance was placed to hold the solution with a bite block for 1 hr in an anesthetization chamber. Mice were then transferred to the operation table and the oral appliance and bite block were removed. In general, there was no remaining solution in the mouth. Mice were returned to the cage in the vivarium. On Day 4 of the MTA, mice were euthanized by 100% CO₂ inhalation and the maxillary tissue was harvested and fixed with 10% buffered formalin for histological evaluation.

Histological analysis

Fixed maxillae were decalcified in 10% EDTA (Sigma–Aldrich) for 3 weeks and then embedded in paraffin. Histological cross-sections were stained with hematoxylin and eosin (HE) and evaluated on a light microscope. Adjacent paraffin sections (4 μ m) were subjected to a heat-induced epitope retrieval procedure and then immunohistochemically stained with polyclonal antibodies to CD45 (#PA5-11671, Thermo Fisher Scientific), COL14A1 (#PA5-49916, Thermo Fisher Scientific), CXCL12 (#PA5-30603, Thermo Fisher Scientific), or Ctsk (PA5-14270, Thermo Fisher Scientific) at the suggested dilution, followed by secondary antibody application, diaminobenzidine staining, and methylene blue counterstaining.

Using maxillary cross-sections of WT, Rag2^{-/-}, and Rag2^{-/-}γc^{-/-} mice, osteoclasts were evaluated by TRAP staining using a commercially available kit (Acid Phosphatase TRAP kit, Sigma–Aldrich), according to the manufacturer's instructions. TRAP-positive cells were counted in the 0.01-mm² area. Statistical analysis to assess the differences among multiple experimental groups was performed using one-way ANOVA with Tukey's multiple-comparison test; $p < 0.05$ was considered to be statistically significant.

Single cell dissociation from mouse maxillary gingiva

On Days 1, 4, and 7 after ligature placement, and on Day 4 of the MTA of CpG ODN or LPS, mice were euthanized by 100% CO₂ inhalation. And maxillary gingival tissues ($n = 4$ per group) were harvested from freshly isolated mouse maxillae.

Collagenase II treatment

The tissues were cut into 1-mm pieces and placed immediately into digestion buffer, containing 1-mg/ml collagenase II (Life Technologies, Thermo Fisher Scientific, Waltham, MA, USA), 10-units/ml DNase I (Sigma–Aldrich, St. Louis, MO, USA), and 1% bovine serum albumin (BSA; Sigma–Aldrich) in Dulbecco's Modified Eagle Medium (DMEM; Life Technologies). The tissues were incubated in digestion buffer for 20 min at 37 °C on a shaker at 150 rpm and then passed through a 70- μ m cell strainer. The collected cells were pelleted at 1,500 rpm for 10 min at 4 °C and resuspended in phosphate-buffered saline (PBS; Life Technologies), supplemented with 0.04% BSA (Cell suspension A).

Trypsin treatment

Immediately following collagenase II treatment, tissues were incubated in 0.25% trypsin (Life Technologies) and 10-units/ml DNase I for 30 min at 37 °C on a 150-rpm shaker. Trypsin was neutralized with fetal bovine serum (Life Technologies), and the tissues were passed through a 70- μ m cell strainer and washed with DMEM. The collected cells were then pelleted at 1,500 rpm for 10 min at 4 °C and resuspended in PBS with 0.04% BSA (Cell suspension B). Cell suspension A and Cell suspension B were combined in one tube. An equal number of combined cell suspensions A and B from four animals per group were combined for scRNA-seq analysis (10X Genomics, San Francisco, CA, USA).

Cell clustering and Identification

Cell Ranger was used to align reads, generate feature–barcode matrices, and perform clustering and gene expression analyses on the scRNA-seq data, and the output from this program was analyzed using the R-program Seurat (<https://satijalab.org/seurat/>). Cells with <2,400 genes detected or >1% mitochondrial gene expression were filtered out as low-quality cells. Individual gene counts for each cell were divided by the total gene counts for that cell and multiplied by a scale factor of 10,000; natural-log transformation was then applied to the counts. The FindVariableFeatures function was used to select 2,000 variable genes with default parameters, and the ScaleData function was used to scale and center the counts in the dataset. Principal

component analysis and Uniform Manifold Approximation and Projection dimensional reduction were performed on variably expressed genes. The cluster markers were found using the FindAllMarkers function, and cell types were manually annotated based on the cluster markers. Cell types were assigned based on expression of cell marker genes, and gene expression within different cell types was displayed using dot plots and violin plots.

Functional annotation and pathway enrichment analysis

Annotation and visualization of GO terms were performed by Metascape (<http://metascape.org/gp/index.html#/main/step1>). The top-100 differentially expressed genes in each population were input and filtered with the term “immune”. Filtered genes were then input, and only “biological process” gene sets were retrieved from the GO database.

Ligand–target matrix prediction

NicheNet (v.1.0.0, <https://github.com/saeyslab/nichenetr>) was used to predict interactions between cell types. In brief, the integrated Seurat object containing each cell subpopulation was input into the NicheNet Seurat wrapper. Sender cells and receiver cells were determined, and interactions between active ligands expressed by sender cells and target receptors expressed by receiver cells were predicted based on information in signaling and ligand–receptor databases.

Acknowledgements

We thank Dr. Connie Lee of Division of Periodontology at UCLA School of Dentistry for her guidance on the clinical manifestations of human periodontitis. We also thank Dr. Yunfeng Li of the Translational Pathology Core Laboratory, Department of Pathology and Laboratory Medicine, David Geffen School of Medicine at UCLA for her immunohistology work. This study was supported by NIH grants R01DE022550, R44DE025524, and C06RR014529 and by SINTX Technologies.

Author contributions

I.N. conceptualized the project and obtained funding. T.K., H.O. and A.H. performed animal studies. H.O. collected micro-CT data from the animal model. T.K. and A.G. performed scRNA-seq and data analysis. T.K. and I.N. drafted the manuscript. All authors gave final approval and agreed to be accountable for all aspects of the work.

Competing interests

I.N. is a consultant for FUJI FILM Corp and BioVinc LLC and received a research fund from SINTX Technologies, Inc. I.N. and A.H. received a research fund from Maruho Co. Ltd. The other authors declare no competing interests.

Data availability statement

We specify where the data supporting the results of this study can be found and provide hyperlinks to publicly archived datasets analyzed or generated during the study, where applicable. scRNA-seq data are found in NIH Gene Expression Omnibus (GSE228635).

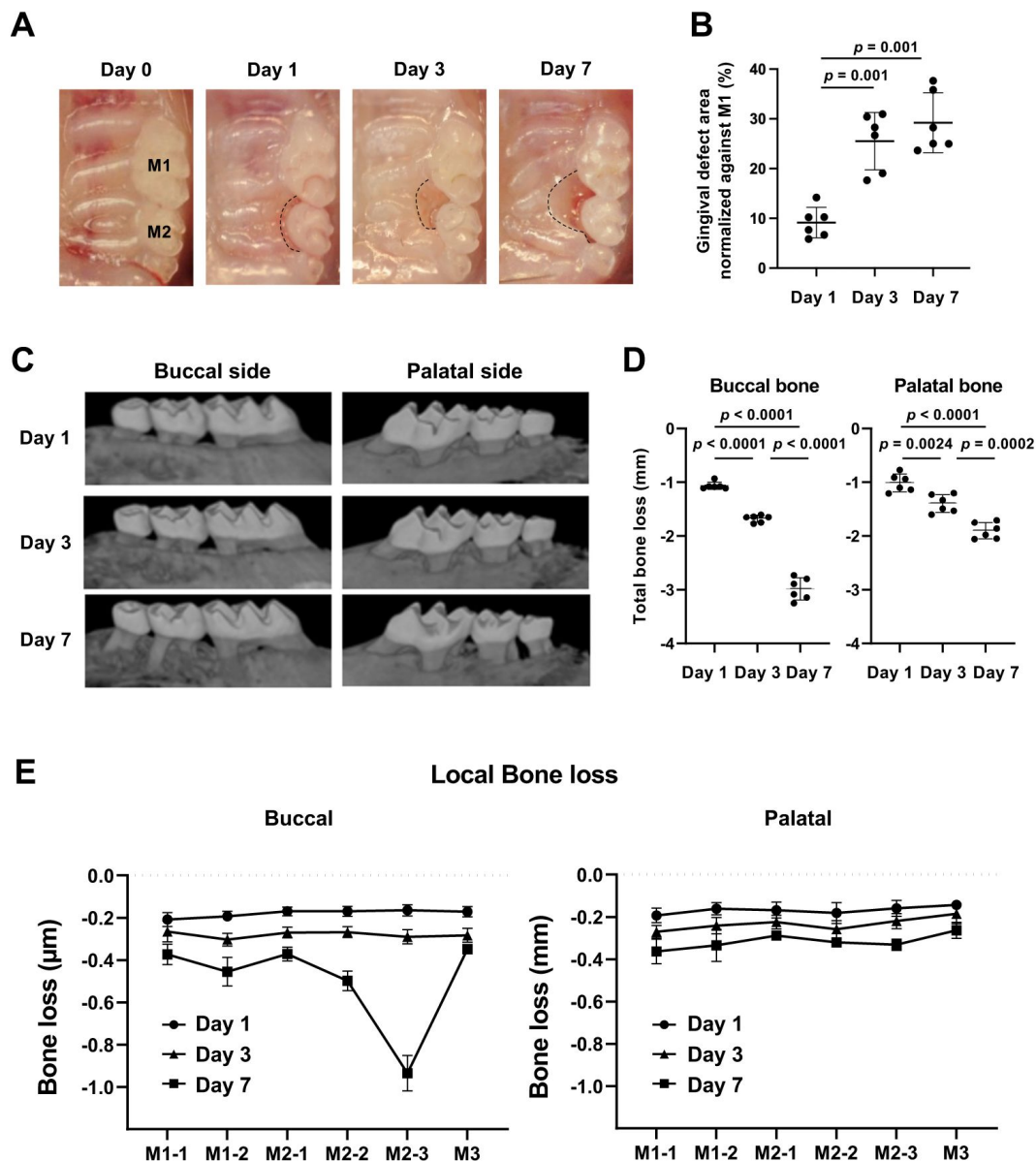


Figure supplement 1.

(A) A ligature (5.0 silk suture) was placed around the maxillary second molar (M2) of wild-type (WT) mice. Representative intra-oral photographs of the maxilla on Day 0, prior to ligature placement (healthy gingiva), and on Days 1, 3, and 7 after ligature placement. (B) The gingival defect area was measured and normalized to the circumferential area of the maxillary first molar (M1) ($n = 6$). Gingival defects appeared on Day 3. (C) Representative micro-computed tomography (microCT) images of the maxilla taken from the lateral view. (D) Alveolar bone loss was determined from the total distance between the cemento-enamel junction (CEJ) and the alveolar bone crest (ABC) of the buccal or palatal bone at six sites in the ligated side ($n = 6$). Alveolar bone loss was apparent on Day 7. (E) Alveolar bone loss was assessed at the mesiobuccal cusp (M1-1), distobuccal cusp (M1-2), and distal cusp (M1-3) of the first molar, the mesiobuccal cusp (M2-1) and distobuccal cusp (M2-2) of the second molar, and the buccal cusp (M3) of the third molar by measuring the distance from the cemento-enamel junction (CEJ) to the alveolar bone crest (ABC) on the buccal or palatal side of the alveolar bone ($n = 6$). Significance was determined by one-way analysis of variance (ANOVA), with Tukey's multiple-comparison test. Data are presented as mean values $\pm$ standard deviation (SD); $p < 0.05$ was considered significant.

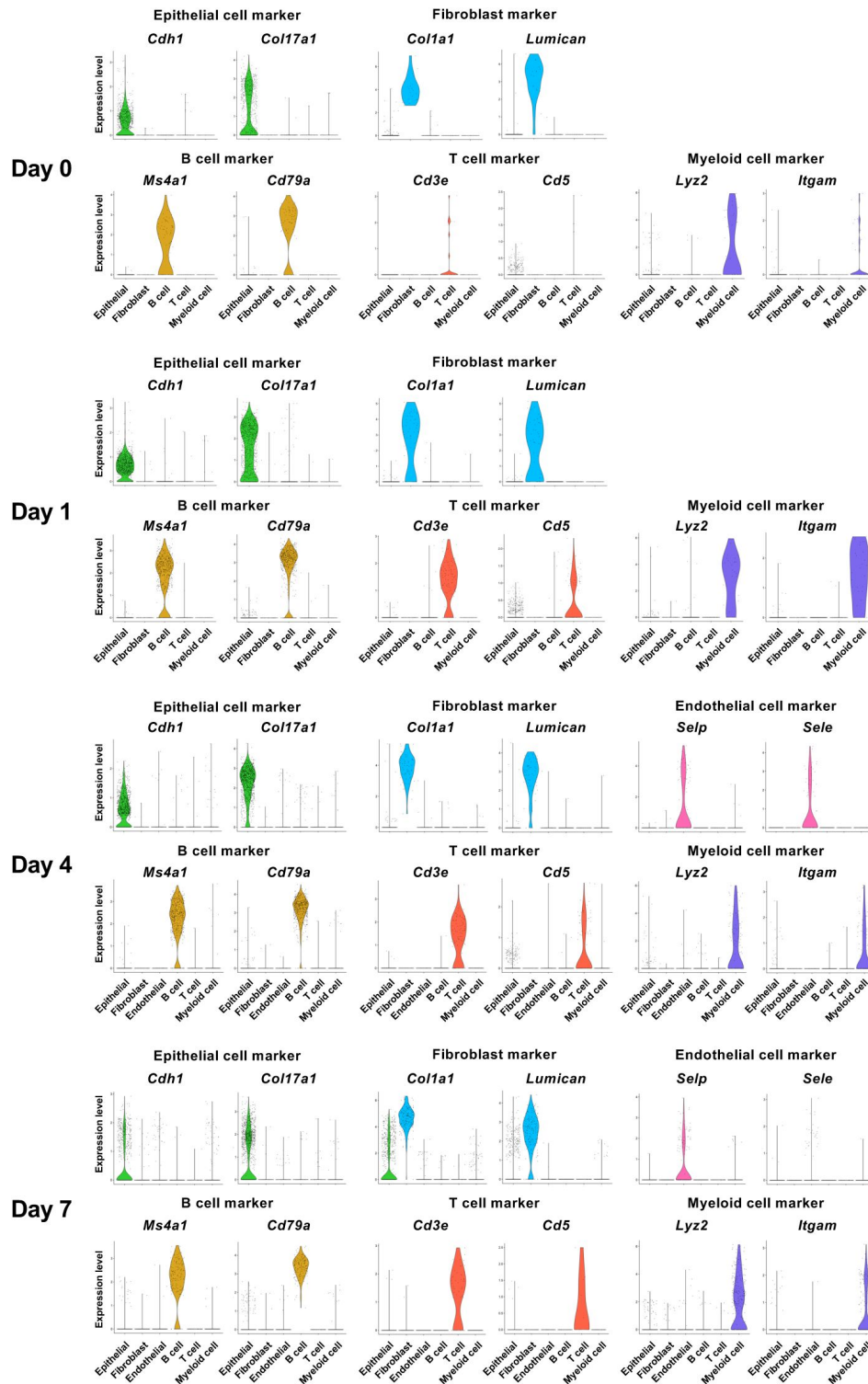


Figure supplement 2.

Identification of major cell types in mouse gingival tissue during periodontitis development by single-cell RNA sequencing (scRNA-seq). Violin plots showing expression levels of cell-type marker genes in each major cell type: epithelial cells, cadherin 1 (*Cdh1*) and type XVII collagen (*Col17a1*); fibroblasts, type I collagen (*Col1a1*) and lumican (*Lum*); endothelial cells, selectin P (*Selp*) and selectin E (*Sele*); B cells, membrane spanning 4 domains A1 (*Ms4a1*) and cluster of differentiation 79A (*Cd79a*); T cells, epsilon subunit of T cell receptor complex (*Cd3e*) and cluster of differentiation 5 (*Cd5*); and myeloid cells, lysozyme 2 (*Lyz2*) and integrin subunit alpha M (*Itgam*).

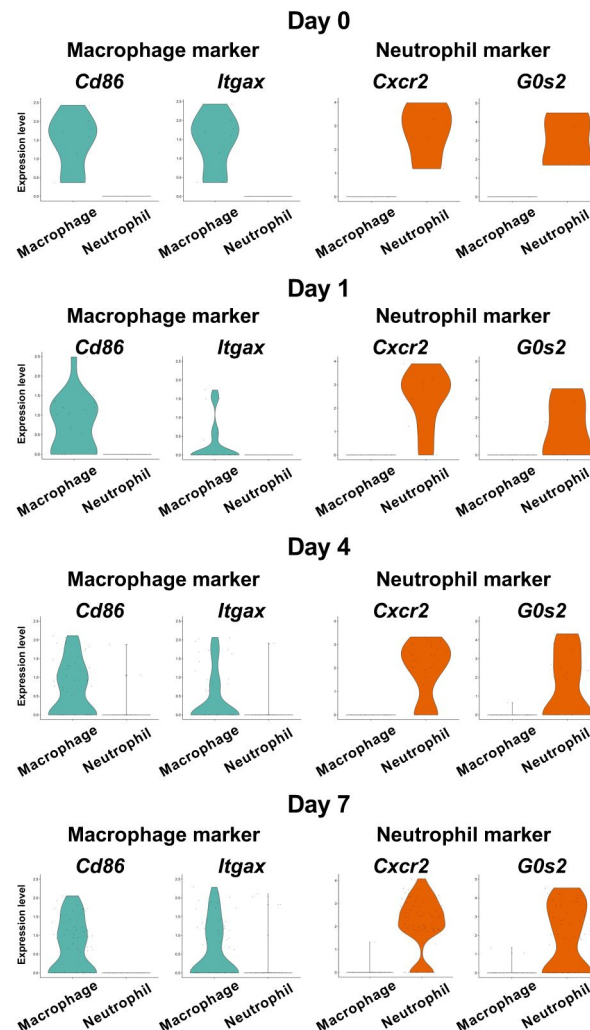


Figure supplement 3.

Identification of myeloid cell subpopulations in mouse gingival tissue during periodontitis development by scRNA-seq. Violin plots showing expression levels of macrophage and neutrophil marker genes in myeloid cell subpopulations: macrophages, cluster of differentiation 86 (*Cd86*) and integrin subunit alpha X (*Itgax*); and neutrophils, CXC motif chemokine receptor 2 (*Cxcr2*) and G0/G1 switch gene 2 (*G0s2*).

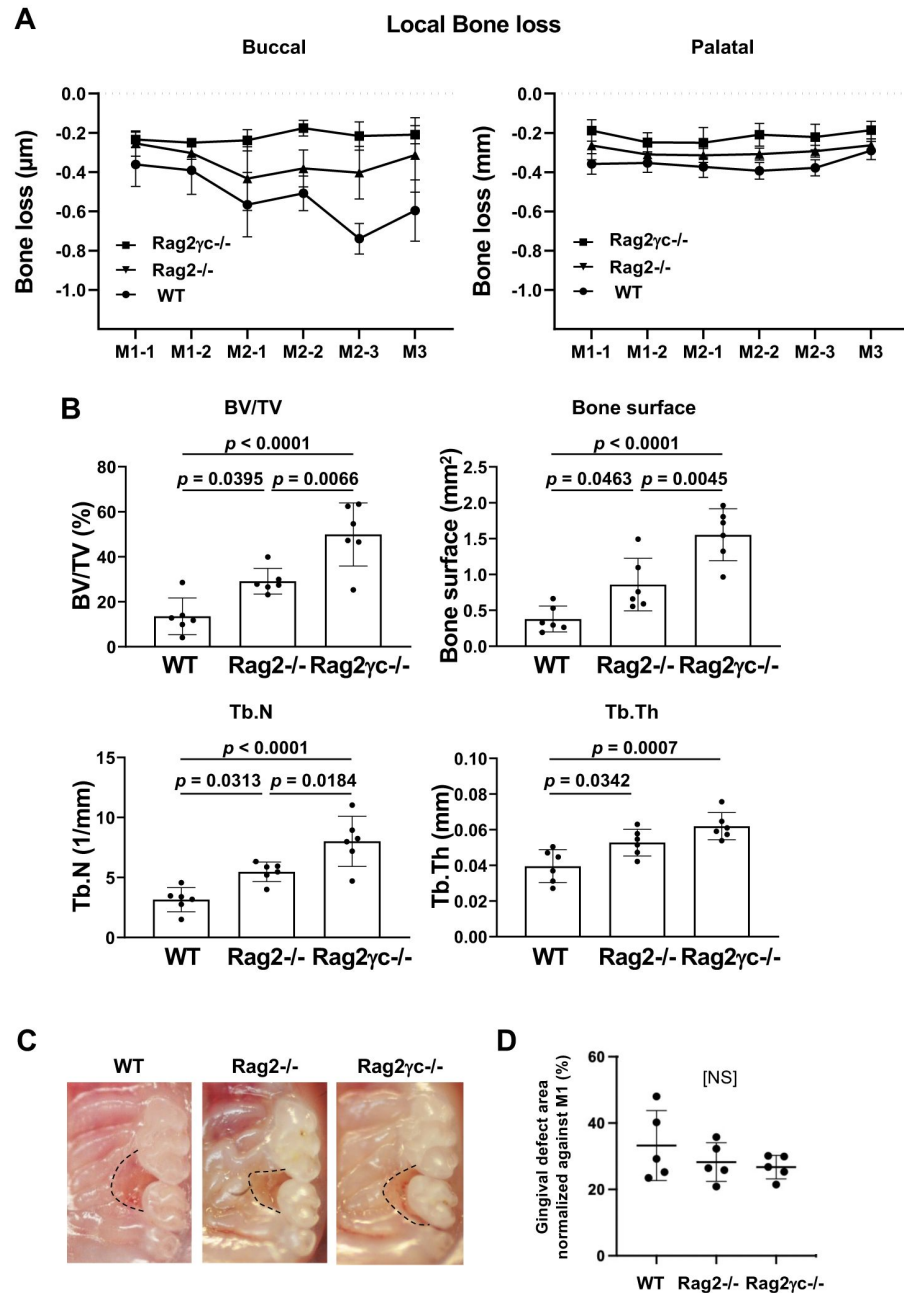
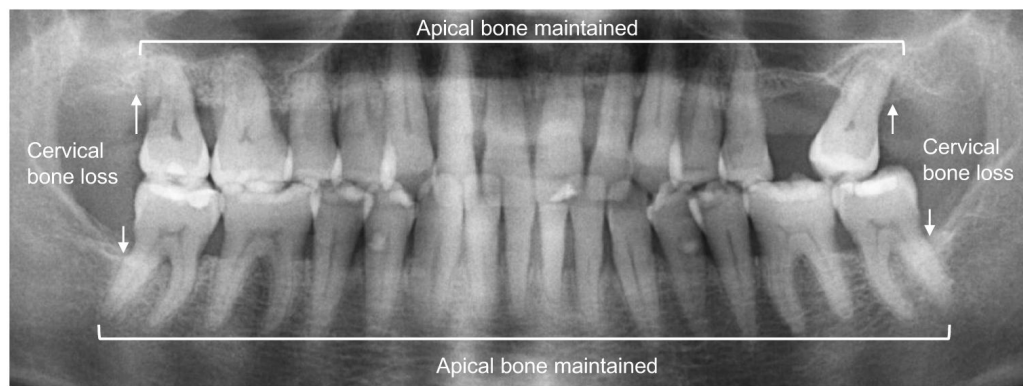


Figure supplement 4.

Effects of innate lymphoid cell (ILC) deletion on alveolar bone loss in the mouse periodontitis model. **(A)** Alveolar bone loss was assessed at six sites by measuring the distance from the CEJ to the ABC on the buccal or palatal side of the alveolar bone of wild-type (WT), Rag2 $^{-/-}$, and Rag2 $^{-/-}$ $\gamma\text{c}^{-/-}$ mice on Day 7 following ligature placement ($n = 6$). **(B)** Bone volume/total volume (BV/TV), bone surface, trabecular number (Tb.N), and trabecular thickness (Tb.Th) in the buccal side of alveolar bone of the second molar were measured on Day 7 ($n = 6$). **(C)** Representative intra-oral photographs of maxilla from WT, Rag2 $^{-/-}$, and Rag2 $^{-/-}$ $\gamma\text{c}^{-/-}$ mice taken on Day 7 following ligature placement. **(D)** The gingival defect area was measured and normalized to the circumferential area of M1 ($n = 5$). Significance was determined by ANOVA, with Tukey's multiple-comparison test. Data are presented as mean values $\pm$ SD; $p < 0.05$ was considered significant. **(E)** Panoramic radiograph of a human patient diagnosed with generalized periodontitis. The alveolar bone resorption was only noted at the cervical alveolar bone (arrows), whereas the apical alveolar bone was well maintained and not affected by periodontitis. **(F)** Periapical dental radiograph and the clinical pictures during the periodontal surgery. A localized alveolar bone loss (arrows) was noted at the cervical periodontal ligament area.

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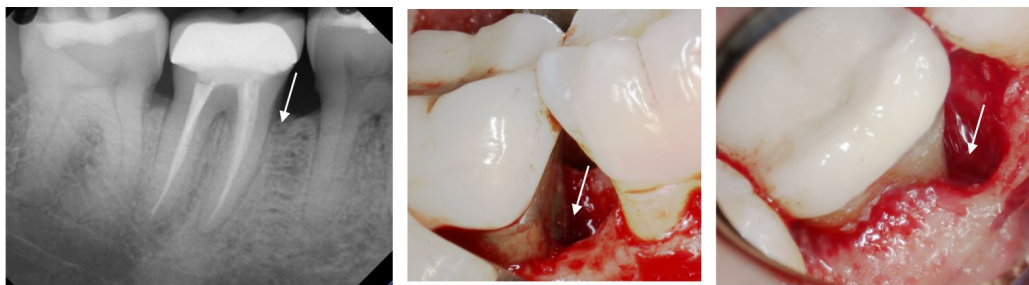


Figure supplement 4. (continued)

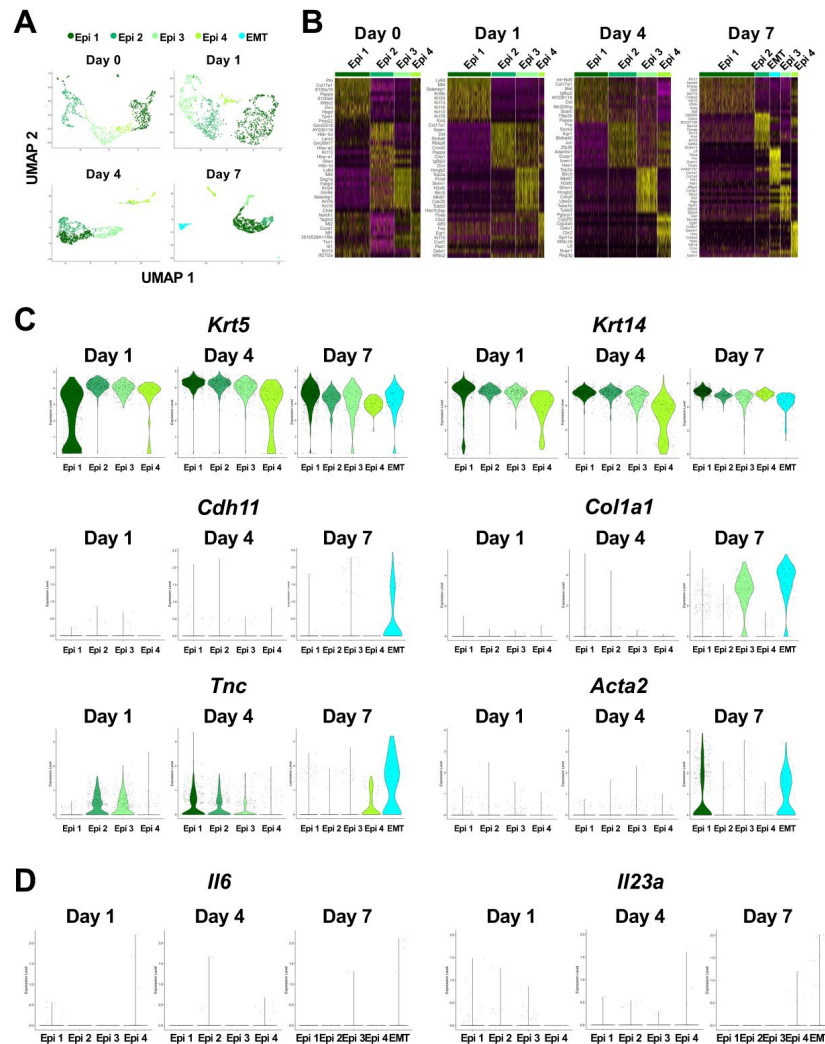


Figure supplement 5.

Epithelial cell subpopulations involved in periodontitis development in mice. **(A)** t-distributed stochastic neighbor embedding (t-SNE) projection plots showing epithelial cell subpopulations in gingival tissue on Days 1, 4, and 7 following ligature placement. Colors indicate cell type, as follows: dark green, epithelial cell population 1 (Epi 1); sea green, epithelial cell population 2 (Epi 2); pale green, epithelial cell population 3 (Epi 3); light green, epithelial cell population 4 (Epi 4); and light blue, epithelial cell population expressing epithelial-mesenchymal transition (EMT) genes. **(B)** Heatmap showing expression of the top-10 differentially expressed genes in Epi 1, Epi 2, Epi 3, Epi 4, and EMT. **(C)** Violin plots showing gene expression levels of keratin 5 (*Krt5*), keratin 14 (*Krt14*), cadherin 11 (*Cdh11*), *Col1a1*, tenascin C (*Tnc*), and smooth muscle aortic actin 2 (*Acta2*) in epithelial cell subpopulations. **(D)** Violin plots showing gene expression levels of interleukin 6 (*Il6*) and interleukin 23A (*Il23a*) in epithelial cell subpopulations.

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Reviewer #1 (Public Review):

In this article, the authors found a distinct fibroblast subpopulation named AG fibroblasts, which are capable of regulating myeloid cells, T cells and ILCs, and proposed that AG fibroblasts function as a previously unrecognized surveillant to orchestrate chronic gingival inflammation in periodontitis. Generally speaking, this article is innovative and interesting.

Reviewer #2 (Public Review):

This study proposed the AG fibroblast-neutrophil-ILC3 axis as a mechanism contributing to pathological inflammation in periodontitis. In this study single-cell transcriptomic analysis was performed. But the signal mechanism behind them was not evaluated.

The authors achieved their aims, and the results partially support their conclusions.

The mouse ligatured periodontitis models differ from clinical periodontitis in human, this study supplies the basis for future research in human.

Author Response

The following is the authors' response to the current reviews.

eLife assessment

The findings of this article provide valuable information on the changes of cell clusters induced by chronic periodontitis. The observation of a new fibroblast subpopulation, named AG fibroblasts, is interesting, and the strength of evidence presented is solid.

We thank the Reviewing Editor and the Senior Editor for the positive assessment and strong support for our study.

Public Reviews:

Reviewer #1 (Public Review):

In this article, the authors found a distinct fibroblast subpopulation named AG fibroblasts, which are capable of regulating myeloid cells, T cells and ILCs, and proposed that AG fibroblasts function as a previously unrecognized surveillant to orchestrate

chronic gingival inflammation in periodontitis. Generally speaking, this article is innovative and interesting.

We truly appreciate this public review.

Reviewer #2 (Public Review):

This study proposed the AG fibroblast-neutrophil-ILC3 axis as a mechanism contributing to pathological inflammation in periodontitis. In this study single-cell transcriptomic analysis was performed. But the signal mechanism behind them was not evaluated.

The authors achieved their aims, and the results partially support their conclusions.

We agree that we must conduct future studies to evaluate our hypothesis.

The mouse ligatured periodontitis models differ from clinical periodontitis in human, this study supplies the basis for future research in human.

This is an important subject. We have previously expressed a concern on the mouse ligature model that the microbial composition of the mouse ligature did not mirror the human oral microbial composition. Therefore, we developed the maxillary topical application (MTA) model, in which human oral biofilm was directly applied to the maxillary gingiva. In this study, the newly developed MTA model was further dissected by single cell RNA seq, which revealed that the extracellular substances of human oral biofilm might be an important trigger of gingival inflammation. RESULT has been revised.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

I appreciate the authors' efforts. I think it would be much better to simplify INTRODUCTION.

INTRODUCTION has been simplified as suggested.

Reviewer #2 (Recommendations For The Authors):

- 1. Many host cells participate in immune responses, such as gingival epithelial cells. AG fibroblast is not the only cell involved in the immune response, and the weight of its role needs to be clarified. So the expression in the conclusion should be appropriate.*

RESPONSE: We agree with this comment. Our study identified the AG fibroblast–neutrophil–ILC3 axis as a previously unrecognized mechanism which could play an additional role in the complex interplay between oral barrier immune cells.

- 1. The main results should be included in the Abstract.*

Abstract has been revised.

The following is the authors' response to the original reviews.

We thank all reviewers for constructive critiques. We plan to perform new experiments and revise our manuscript accordingly. The text and Figures are currently undergoing the revision process. Below highlights our revision plan.

eLife assessment

The findings of this article provide valuable information on the changes of cell clusters induced by chronic periodontitis. The observation of a new fibroblast subpopulation, which was named as AG fibroblasts, was quite interesting, but needs further evidence. The strength of evidence presented is incomplete.

We discovered a new subpopulation of gingival fibroblasts, named AG fibroblasts, using non-biased single cell RNA sequencing (scRNA-seq) of mouse gingival samples undergoing the development of ligature-induced periodontitis. AG fibroblasts exhibited a unique gene expression profile: [1] constitutive expression of type XIV collagen; and [2] ligature-induced upregulation of Toll-Like Receptors and their downstream signals as well as chemokines such as CXCL12. Thus, we have hypothesized that AG fibroblasts initially sense the pathological stress including oral microbial stimuli and secrete inflammatory signals through chemokine expression.

The current manuscript examined the relationship between AG fibroblasts and oral barrier immune cells focusing on the chemokines and other ligands derived from AG fibroblasts and their putative receptors in those immune cells. Using scRNA-seq data mining programs, our data demonstrated the compelling evidence that AG fibroblasts should play a critical role in orchestrating the oral barrier immunity, at least at the early stages of periodontal inflammation.

We agree that it is important to explore the functional/pathological role of AG fibroblasts. In this revision, we further investigated the role of TLRs in the pathogen sensing mechanism of AG fibroblasts. To accomplish this goal, we applied a newly developed mouse model in which mice were exposed to the maxillary topical application (MTA) of oral microbial pathogens without the ligature placement. With 1 hr exposure with human oral biofilm, not with planktonic microbiota, the mice maxillary tissue exhibited measurable degradation as evidenced by the activation of cathepsin K. To dissect the role of TLRs, we applied the putative stimulants of TLR9 and TLR2/4 using the discrete MTA model. The scRNA-seq from the MTA model revealed that the application of unmethylated CpG oligonucleotide and *P. gingivalis* lipopolysaccharide (LPS), respectively, induced the activation of chemokines by AG fibroblast.

The revised manuscript reported this critical data with the detailed information. As such the additional figures and corresponding results, discussion and materials & methods were included.

Public Reviews:

Reviewer #1 (Public Review):

In this article, the authors found a distinct fibroblast subpopulation named AG fibroblasts, which are capable of regulating myeloid cells, T cells and ILCs, and proposed that AG fibroblasts function as a previously unrecognized surveillant to orchestrate chronic gingival inflammation in periodontitis. Generally speaking, this article is innovative and interesting, however, there are some problems that need to be addressed to improve the quality of the manuscript.

We appreciate this comment. As suggested, we further investigated the surveillant function of AG fibroblasts by reanalyzing the scRNA-seq data for stress sensing receptors such as Toll-Like Receptors (TLR). In the revision, we addressed the role of TLR in the activation of AG fibroblasts using a newly developed mouse model employing the maxillary topical application (MTA) of putative TLR stimulants. The new information clearly demonstrated that

AG fibroblasts play a pivotal role as the surveillant and translating the pathogenic stimulants to oral barrier inflammation through chemokine expression.

Reviewer #2 (Public Review):

This study proposed the AG fibroblast-neutrophil-ILC3 axis as a mechanism contributing to pathological inflammation in periodontitis. However, the immune response in the vivo is very complex. It is difficult to determine which is the cause and which is the result. This study explores the relevant issue from one dimension, which is of great significance for a deeper understanding of the pathogenesis of periodontitis. It should be fully discussed.

We appreciate this comment. We expanded the current understanding of oral immune signal communication in Discussion and highlight how AG fibroblast may fit to it. To address this question, we expanded our investigation in the pathological signal detection by AG fibroblasts by employing the newly developed maxillary topical application (MTA) model. The revised manuscript contains the new information and expanded the discussion in the context of complex immune response.

Reviewer #1 (Recommendations For The Authors):

Detailed comments are listed below:

Abstract :

I am confused about the expression of "human periodontitis-like phenotype". How does the authors define this concept? Periodontitis is a complex disease, despite that alveolar bone resorption is a typical manifestation of periodontitis, its characteristics remain to be further studied. I hope the authors can provide some detailed information about this concept or describe it in another way.

This is an important comment. Radiographically, human periodontitis is diagnosed by alveolar bone resorption from the cervical region, not from root apex. To highlight this, we present dental radiographs of human periodontitis as supplementary information. However, we agree with this comment, our statement should be limited to alveolar bone resorption pattern in Rag2KO and Rag2gcKO mice. Abstract be revised accordingly.

Introduction:

It is recommended to simplify the first to third paragraphs, and briefly explain the functions of various types of cells in different stages of periodontitis, as well as the role of different cluster markers play across the time course of periodontal inflammation development.

Following this recommendation, INTRODUCTION has been simplified.

Results:

- 1. It is recommended to add HE staining and immunohistochemistry staining to observe the inflammation, tissue damage, and repair status from 0 to 7 days, so that readers can understand cell phenotype changes corresponding to the periodontitis stage. The observation index can include inflammation and vascular related indicators.*

As recommended, representative histological figures were included. We further performed new immunohistochemistry experiment of mouse gingival tissue (D0, D1, D3, D7) highlighting the infiltration of CD45+ immune cells. We found that inflammatory vascular formation in the H&E histology, which was highlighted. To characterize the tissue damage, the histological

sections were stained by picrosirius red to highlight the change in collagen connective tissue of PDL and gingiva.

1. Figure 1A-1D can be placed in the supplementary figure.

Combining the new data above, Figure 1 was revised as suggested.

1. I suggest the authors to put the detection of the existence of AG fibroblasts before exploring its relationship with other types of cells.

1. The layout of the picture should be closely related to the topic of the article. It is recommended to readjust the layout of the picture. Figure 1 should be the detection of AG cells and their proportion changes from 0 to 7 days. In other figures, the authors can separately describe the proportion changes of myeloid cells, T cells and ILCs, and explored the association between AG fibroblasts and these cell types.

As suggested, the presentation order of Figures and text was revised to bring the information about AG fibroblasts first. The chemokine-receptor analysis was moved below.

1. Please provide the complete form of "KT" in Line 162.

KT fibroblasts (fibroblasts keeping typical phenotype) was described in the text.

Methods:

It is recommended to separately list the statistical methods section. The statistical method used in the article should be one-way ANOVA.

A separate statistical method section is created. As pointed out, we used one-way ANOVA with post-hoc Tukey test (when multiple groups were compared).

Discussion:

I suggest the authors remove Figures 3-6 from the discussion section. For example, in Line 283, "(Figure 3 and 4)" should be removed.

Revised as suggested.

Reference:

Some information for the references is missing. For example, "Lin P, et al. Application of Ligature-Induced Periodontitis in Mice to Explore the Molecular Mechanism of Periodontal Disease. Int J Mol Sci 22, (2021)" should be "Lin P, et al. Application of Ligature-Induced Periodontitis in Mice to Explore the Molecular Mechanism of Periodontal Disease. Int J Mol Sci 22, 8900 (2021)". It is necessary to recheck all references.

The reference has been checked for the accuracy and the omission pointed out was corrected. Although we used EndNote program, we found some more inaccuracy in the references that were manually corrected. We appreciate your suggestion.

Reviewer #2 (Recommendations For The Authors):

- 1. Many host cells participate in immune responses, such as gingival epithelial cells. AG fibroblast is not the only cell involved in the immune response, and the weight of its role needs to be clarified. So the expression in the conclusion should be appropriate.*

Following this critique, we revised INTRODUCTION, DISCUSSION and CONCLUSION, to highlight how AG fibroblasts function within a comprehensive immune response network.

- 1. This study cannot directly answer the issue of the relationship between periodontitis and systemic diseases.*

We agree with this critique. We either deleted or de-emphasized the relationship between periodontitis and systemic diseases throughout the text.